Topology

Math 817

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Warning!

Proceed with caution. These notes are under construction and are 100% guaranteed to contain typos. If you find any typos or errors, we will be most grateful to you for letting me know.

Acknowledgements

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Chapter 1

Topological spaces and continuous functions

1.1 Topological spaces

Definition 1.1.1. Let X be a set. A **topology** on X is a collection $\mathcal{T}$ of subsets of X, called **open sets**, satisfying the following properties:

- The empty set belongs to $\mathcal{T}$: $\emptyset \in \mathcal{T}$.
- The entire space X belongs to $\mathcal{T}$: $X \in \mathcal{T}$.
- If U and V are in $\mathcal{T}$, then so is $U \cap V$.
- If $\{U_\alpha\}_{\alpha \in I}$ is any collection of open sets in $\mathcal{T}$, then $\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T}$.

A **topological space** is a pair $(X, \mathcal{T})$ where $\mathcal{T}$ is a topology on the set $\mathcal{T}$.

We will often refer to an element $x \in X$ in a topological space $(X, \mathcal{T}_X)$ as a **point**.

Remark 1.1.2. We may sometimes abuse notation and say "Let X be a topological space", which should be interpreted as "Let X be a set equipped with a topology $\mathcal{T}$". Ironically, the topology $\mathcal{T}$ (meaning, the collection of open sets) is a lot more important than the set: given $\mathcal{T}$, we can recover the set X by taking the union of all the sets in $\mathcal{T}$.

Remark 1.1.3. Let $(X, \mathcal{T})$ be a topological space. Given any *finite* collection $U_1, \dots, U_n$ of open sets, applying the intersection axiom repeatedly we get that the intersection $U_1 \cap \cdots \cap U_n$ is also an open set. However, note that the arbitrary intersection of open sets might not be an open set.

Definition 1.1.4. Let X be any set. The **trivial topology**, also known as the **indiscrete topology** on X, is the topology $\mathcal{T} = \{\emptyset, X\}$ consisting of only the empty set and X. The **discrete topology** on X is the topology $\mathcal{T} = \mathcal{P}(X)$ consisting of all the subsets of X.

One of the most important examples of a topology is the familiar standard topology on $\mathbb{R}$, or more generally on $\mathbb{R}^n$, which one may take as inspiration for everything we will do.

Notation 1.1.5. Recall that for real numbers $a < b$, by (a, b) we mean the set

$$(a, b) := \{r \in \mathbb{R} \mid a < r < b\},$$

also called the **open interval** from a to b. Similarly,

$$[a, b] := \{r \in \mathbb{R} \mid a \leqslant r \leqslant b\}$$

is the **closed interval** from a to b. We may also write $[a, b)$ or $(a, b]$ for the intervals that include a but not b, or b but not a, respectively.

Example 1.1.6. The **Euclidean topology** or **standard topology** on the set of real numbers $\mathbb{R}$ is defined as follows: a subset U of $\mathbb{R}$ is *open* if for all $x \in U$ there is a positive real number $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq U$. Note that every nonempty open set in this topology is infinite.

In more informal terms, U is open if and only if every point in U is a strictly positive distance away from all points not in U. In particular, for all real numbers $a < b$, the set (a, b) is an open set in the standard topology for $\mathbb{R}$. Let us prove this carefully.

Let $x \in (a, b)$. Note that $a < x < b$, so $\varepsilon := \min\{b - x, x - a\} > 0$. We claim that $(x - \varepsilon, x + \varepsilon) \subseteq (a, b)$. To see this, pick $r \in (x - \varepsilon, x + \varepsilon)$. Since $\varepsilon \leqslant x - a$, we have $x - \varepsilon \geqslant a$, and hence $r > x - \varepsilon \geqslant a$ gives $r > a$. A similar argument shows that since $\varepsilon \leqslant b - x$, we have $r < b$. This proves that $r \in (a, b)$. Therefore, since r was arbitrary, we have $(x - \varepsilon, x + \varepsilon) \subseteq (a, b)$. Since this holds for any $x \in (a, b)$, this shows that (a, b) is open.

Remark 1.1.7. Note that while the standard topology on $\mathbb{R}$ is the one that most matches our natural intuition, the set $\mathbb{R}$ can be given other topologies; for example, we can consider the discrete topology or the trivial topology on $\mathbb{R}$. In fact, given any set X, the term *open subset* depends on the topology we are considering, and it is not in any sense intrinsic to X.

But of course, the standard topology plays a special role in many other areas of mathematics, so it is worth keeping it in mind for some guidance and intuition. In fact, there is also a standard topology on $\mathbb{R}^n$:

Example 1.1.8. The **Euclidean topology** or **standard topology** on $X = \mathbb{R}^n$ is given as follows: a subset U of X is open if for every $v \in U$ there exists a positive real number $\varepsilon > 0$ such that $B_\varepsilon(v) \subseteq U$, where

$$B_\varepsilon(v) = \{w \in \mathbb{R}^n \mid \operatorname{dist}(v, w) < \varepsilon\}$$

and dist denotes the Euclidean distance between v and w. In other words, $B_\varepsilon(v)$ denotes the open ball of radius ε centered at v. One can adapt the proof we gave in Example 1.1.8 to show that this is indeed a topology. Note in particular that every nonempty open set in this topology is infinite.

Example 1.1.9. We can now give an explicit example showing that the arbitrary intersection of open sets if not necessarily open. In $\mathbb{R}$ with the standard topology, the intersection of open sets

$$\bigcap_{n \geqslant 1} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$$

cannot be an open set, since it is finite.

Example 1.1.10. Let $X = \{a, b, c\}$ be an element with three elements, and consider

$$\mathcal{T} = \{\emptyset, X, \{a, b\}, \{b\}, \{b, c\}\}.$$

One can check (however tedious that may be) is that $\mathcal{T}$ is in fact a topology on X.

Example 1.1.11 (The Finite Complement Topology)**.** Let X be any set. The **finite complement topology** on X is the topology

$$\mathcal{T} = \{U \subseteq X \mid X \setminus U \text{ is finite}\} \cup \{\emptyset\}$$

where the open sets are the empty set and all the sets that contain all but a finite number of elements of X.

Exercise 1.1.12. Check that the finite complement topology is indeed a topology.

We will often be concerned with topologies on $\mathbb{R}$ or $\mathbb{R}^n$, so we will make use of some of the nice properties of the real numbers. One such property we will use is the **Completeness Axiom** of the real numbers:

Every nonempty, bounded above subset of $\mathbb{R}$ *has a least upper bound.*

Let us properly recall what all these terms mean:

Definition 1.1.13. A subset $X \subseteq \mathbb{R}$ is **bounded above** if there exists $u \in \mathbb{R}$ such that

$$x \leqslant u \quad \text{for all } x \in X.$$

Any u with this property is an **upper bound** for X. A subset $X \subseteq \mathbb{R}$ is **bounded below** if there exists $\ell \in \mathbb{R}$ such that

$$\ell \leqslant x \quad \text{for all } x \in X.$$

Any ℓ with this property is a **lower bound** for X. A **least upper bound** for a subset $X \subseteq \mathbb{R}$ is a real number $u \in \mathbb{R}$ such that u is an upper bound for X and $u \leqslant w$ for all upper bounds w for X. A **greatest lower bound** for a subset $X \subseteq \mathbb{R}$ is a real number $\ell \in \mathbb{R}$ such that ℓ is an upper bound for X and $w \leqslant \ell$ for all lower bounds w for X.

Example 1.1.14. The **infinite ray topology** on $\mathbb{R}$ is given by

$$\mathcal{T} = \{(-\infty, a) \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}.$$

We can think of $\emptyset$ and $\mathbb{R}$ as informally arising by taking $a = -\infty$ and $a = \infty$, respectively, though this is not formally correct (since these are not elements of $\mathbb{R}$).

Exercise 1.1.15. Show that the infinite ray topology on $\mathbb{R}$ is indeed a topology.
Hint: you may need to use the Completeness Axiom for $\mathbb{R}$.

Exercise 1.1.16. Is the following a topology on $\mathbb{R}$?

$$\mathcal{T} = \{(-\infty, a] \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$$

Example 1.1.17 (The line with two origins)**.** Let L be the real line with two rival origins; more precisely, we add in a point to the real line, and call our two origins 0_1 and 0_2:

$$L = (\mathbb{R} \setminus \{0\}) \cup \{\mathbf{0}_1, \mathbf{0}_2\}.$$

If one removes $\mathbf{0}_2$ from L, we get the set $\mathbb{R}_1 = (\mathbb{R} \setminus \{0\}) \cup \{\mathbf{0}_1\} = \mathbb{R} \setminus \{\mathbf{0}_2\}$, which will identify with $\mathbb{R}$ itself in the natural way, by identifying $\mathbf{0}_1$ with 0. Similarly, we may identify $\mathbb{R}_2 = X \setminus \{\mathbf{0}_1\} = (\mathbb{R} \setminus \{0\}) \cup \{\mathbf{0}_2\}$ with $\mathbb{R}$. The **line with two origins** is the topological space with underlying set L where a subset U of L is open if $U \setminus \{\mathbf{0}_2\}$ is an open subset of $\mathbb{R}_1$ equipped with the standard topology and $U \setminus \{\mathbf{0}_1\}$ is an open subset of $\mathbb{R}_2$ equipped with the standard topology. We claim that the line with two origins forms a topology on X, which we leave as an exercise.

Let U be any open set in L containing 0_1. Then $U \cap \mathbb{R}_1$ is an open set in the standard topology of $\mathbb{R}$, so there exists $\varepsilon > 0$ such that $(-\varepsilon, \varepsilon) \subseteq U \cap \mathbb{R}_1$. Then in L,

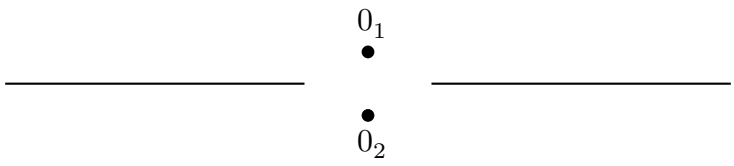
$$(-\varepsilon, 0) \cup \{\mathbf{0}_1\} \cup (0, \varepsilon) \subseteq U.$$

Analogously, for any open set V in L containing 0_2 there exists $\delta > 0$ such that

$$(-\delta, 0) \cup \{\mathbf{0}_2\} \cup (0, \delta) \subseteq V.$$

In particular, this shows that if $U \ni \mathbf{0}_1$ and $V \ni \mathbf{0}_2$ are open subsets of L, then $U \cap V$ is nonempty. That is, one cannot *separate* the two origins from each other by open subsets. We will come back to this point later and understand it better.

Here is a depiction of the line with two origins:



Exercise 1.1.18. Show that the line with two origins is a topological space.

Here are a few other general constructions that produce a topology for any set:

Example 1.1.19. Let X be a nonempty set and fix a point $a \in X$. The **included point topology** on X with included point a is the topology given by

$$\mathcal{T} = \{U \subseteq X \mid a \in U\} \cup \{\emptyset\}.$$

The **excluded point topology** on X with excluded point a is the topology

$$\mathcal{T} = \{U \subseteq X \mid a \notin U\} \cup \{X\}.$$

Exercise 1.1.20. Let $(X, \mathcal{T})$ be a topological space, and let A be a subset of X. Suppose that for every $x \in A$, there exists an open set U such that $x \in U \subseteq A$. Show that A is an open set.

Definition 1.1.21. Let $(X, \mathcal{T}_X)$ be a topological space. An **open neighborhood**, or simply **neighborhood**, of $x \in X$ is any open subset U such that $x \in U$.

Many statements in topology are best phrased in terms of neighborhoods, as we will soon see.

Exercise 1.1.22. Let X be any nonempty set and fix a point $a \in X$. Show that the included point topology and the excluded point topology on X with point a are both in fact topologies.

Given two different topologies on the same set, we may compare them.

Definition 1.1.23. Let X be a set and let $\mathcal{T}$ and $\mathcal{T}'$ be two topologies on X. We say that $\mathcal{T}$ is **finer** than $\mathcal{T}'$, or equivalently that $\mathcal{T}'$ is **coarser** than $\mathcal{T}$, if $\mathcal{T}' \subseteq \mathcal{T}$.

Example 1.1.24. Given any set X, the trivial topology is coarser than the discrete topology, and the discrete topology is finer than the trivial topology.

Example 1.1.25. In $\mathbb{R}$, the trivial topology is coarser than the standard topology, and the discrete topology is finer than the standard topology.

Remark 1.1.26. Note that two topologies $\mathcal{T}_1$ and $\mathcal{T}_2$ can also be **incomparable**, meaning that $\mathcal{T}_1 \not\subseteq \mathcal{T}_2$ and $\mathcal{T}_2 \not\subseteq \mathcal{T}_1$. More explicitly, there is an open set $U \in \mathcal{T}_1$ with $U \notin \mathcal{T}_2$ and there is an open set $V \in \mathcal{T}_2$ with $V \notin \mathcal{T}_1$.

Problem 1.1.27. Compare the following topologies on $\mathbb{R}$:

a) $\mathcal{T}_1$ the standard topology.

b) $\mathcal{T}_2$ the finite complement topology.

c) $\mathcal{T}_3 = \{(-\infty, a) \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$ the infinite ray topology.

d) $\mathcal{T}_4 = \{(a, \infty) \mid a \in \mathbb{R}\} \cup \{\emptyset, \mathbb{R}\}$.

e) $\mathcal{T}_5$ the discrete topology.

f) $\mathcal{T}_6$ the indiscrete topology.

Exercise 1.1.28. Let $\{\mathcal{T}_\alpha\}_{\alpha \in I}$ be a nonempty collection of topologies on a set X. Show that

$$\bigcap_{\alpha \in I} \mathcal{T}_\alpha$$

is a topology on X.

Definition 1.1.29. Let $(X, \mathcal{T})$ be a topological space. An **open cover** for $(X, \mathcal{T})$ is any collection $\{U_\alpha\}_{\alpha \in I}$ of open sets such that

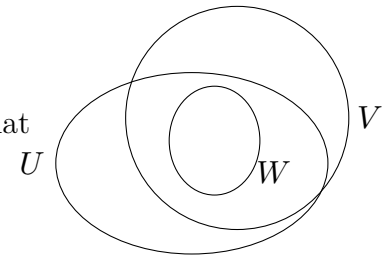
$$\bigcup_{\alpha \in I} U_\alpha = X.$$

1.2 Bases

A basis for a topology on X is a collection of subsets of X that naturally determines a topology.

Definition 1.2.1. Let X be a set. A **basis** for a topology $\mathcal{T}$ on X consists of a collection of subsets of X, $\mathscr{B} \subseteq \mathcal{P}(X)$, with the following properties:

- For every $x \in X$, there exists $B \in \mathscr{B}$ such that $x \in B$. Equivalently,
$$\bigcup_{B \in \mathscr{B}} B = X.$$
- For all $U, V \in \mathscr{B}$, if $x \in U \cap V$, there exists $W \in \mathscr{B}$ such that
$$x \in W \subseteq U \cap V.$$



Remark 1.2.2. Note that the second condition in the definition of a basis is automatically satisfied if the intersection of two basis elements is again in the basis. However, as we will soon see, that is not what happens in general.

Definition 1.2.3. Given a basis $\mathscr{B}$ for a topology on X, the **topology generated by** $\mathscr{B}$ is the topology whose open sets are the subsets $U \subseteq X$ with the following property: for every $x \in U$, there exists $B \in \mathscr{B}$ such that $x \in B \subseteq U$.

Remark 1.2.4. Note that if $\mathcal{T}$ is the topology generated by the basis $\mathscr{B}$, each element $B \in \mathscr{B}$ is itself an open set in $\mathcal{T}$.

Lemma 1.2.5. *Let X be a set and $\mathscr{B}$ be a basis for a topology on X. Then the topology generated by $\mathscr{B}$ is indeed a topology on X.*

Proof. Let $\mathcal{T}$ be the topology generated by $\mathscr{B}$. The condition for $\emptyset \in \mathcal{T}$ is vacuously satisfied. Moreover, by definition of basis, for every $x \in X$ there exists $U \in \mathscr{B}$ such that $x \in U (\subseteq X)$, so $X \in \mathcal{T}$ by definition of $\mathcal{T}$.

Suppose that $A, B \in \mathcal{T}$, and let us show that $A \cap B \in \mathcal{T}$. Pick any $x \in A \cap B$. Then since $A \in \mathcal{T}$, by definition there exists $U \in \mathscr{B}$ such that $x \in U \subseteq A$ and, since $B \in \mathcal{T}$, there exists $V \in \mathscr{B}$ such that $x \in V \subseteq B$. By definition of a basis, there exists $W \in \mathscr{B}$ such that $x \in W \subseteq U \cap V \subseteq A \cap B$. This shows that $A \cap B \in \mathcal{T}$.

Consider an arbitrary collection of $U_\alpha \in \mathcal{T}$ indexed by a set I. Set
$$U = \bigcup_{\alpha \in I} U_\alpha.$$
To show that $U \in \mathcal{T}$, consider $x \in U$. Then $x \in U_\alpha$ for some $\alpha \in I$, and since $U_\alpha \in \mathcal{T}$, there exists $B_\alpha \in \mathscr{B}$ such that $x \in B_\alpha \subseteq U_\alpha \subseteq U$. Since this holds for all $x \in U$, we have shown that $U \in \mathcal{T}$. □

Definition 1.2.6. Given a basis $\mathscr{B}$ for a topology on X, the elements of $\mathscr{B}$ are called **basic open sets**.

Definition 1.2.7. Let $(X, \mathcal{T})$ be a topological space. A **basis** for the topology $\mathcal{T}$ on X is a basis for a topology on X that generates the topology $\mathcal{T}$.

Remark 1.2.8. Every topology has a basis. Indeed, if $(X, \mathcal{T})$ is a topological space, then $\mathcal{T}$ is a basis for itself.

However, we may be interested in smaller, more tractable bases. Once we have a basis for a given topology, we can reconstruct all open sets from the basic open sets: each open set is in fact a union of basic open sets.

Theorem 1.2.9. *Let $\mathscr{B}$ be a basis for a topology on a set X and let $\mathcal{T}$ be the topology generated by $\mathscr{B}$. A subset U of X is open for the topology $\mathcal{T}$ if and only if U is an arbitrary union of basic open sets in $\mathscr{B}$.*

Proof. ($\Leftarrow$) By definition, each member of $\mathscr{B}$ is open (meaning it belongs to $\mathcal{T}$). Therefore, by definition of topology any arbitrary union of basic open sets in $\mathscr{B}$ must be open.

($\Rightarrow$) Let U be open in the topology $\mathcal{T}$. By definition of the topology $\mathcal{T}$ generated by $\mathscr{B}$, for each $x \in U$ there is a basic open $B_x \in \mathscr{B}$ such that $x \in B_x \subseteq U$. Therefore,

$$U = \bigcup_{x \in U} B_x.$$

□

Example 1.2.10. Let $\mathscr{B}$ be the collection of nonempty open intervals of $\mathbb{R}$:

$$\mathscr{B} := \{(a, b) \mid a < b\}.$$

We claim that $\mathscr{B}$ is a basis for a topology on $\mathbb{R}$. First, given any $x \in \mathbb{R}$ we can always find $a, b \in \mathbb{R}$ such that $a < x < b$, so $x \in (a, b)$. Moreover, given any two (a, b) and (c, d) in $\mathscr{B}$, if $x \in (a, b) \cap (c, d)$ then $c < x < b$ and in particular $c < b$. Similarly, $a < d$. Set

$$r = \max\{a, c\} \qquad \text{and} \qquad s = \min\{b, d\}.$$

Note that $r < x < s$. Then $x \in (r, s) \subseteq (a, b) \cap (c, d)$.

Remark 1.2.11. Example 1.2.10 is misleadingly simple, since in this case $(a, b) \cap (c, d)$ is another open interval whenever it is nonempty. In general, it need not be the case that the intersection of any two members basis open sets is either empty or a member of the basis, as we will see in Example 1.2.12.

Example 1.2.12. In $\mathbb{R}^n$, the collection of all open balls

$$B_\varepsilon(v) = \{w \in \mathbb{R}^n \mid \operatorname{dist}(v, w) < \varepsilon\}$$

with all centers and all radii $\varepsilon > 0$ forms a basis for the standard topology. When $n = 1$, note that this is precisely the basis we gave in Example 1.2.10. When $n \geqslant 2$, this is a bit more interesting. In particular, note that two balls with different centers can have a nontrivial intersection which is no longer a ball.

Note that the same topological space may have different bases.

Example 1.2.13. We claim that the collection

$$\mathscr{B} = \{(a, b) \mid a, b \in \mathbb{Q}\}$$

of open intervals *with rational endpoints* is a basis for the standard topology on $\mathbb{R}$. First, one can repeat the argument in Example 1.2.10 to show that $\mathscr{B}$ is indeed a basis for a topology on $\mathbb{R}$. Note in particular that this basis $\mathscr{B}$ is contained in the basis for the standard topology we gave in Example 1.2.10, so the topology generated by $\mathscr{B}$ is necessarily contained in the standard topology. It thus suffices to show that if U is an open set in the standard topology, then U is open in the topology generated by $\mathscr{B}$.

Let $x \in U$. By definition, there exists $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \subseteq U$. By the Density of the Rationals in $\mathbb{R}$, we can find rational numbers $q, r \in \mathbb{Q}$ such that $x - \varepsilon < q < x$ and $x < r < x + \varepsilon$. Therefore, $x \in (q, r) \in \mathscr{B}$. Note that in fact

$$x \in (q, r) \subseteq (x - \varepsilon, x + \varepsilon) \subseteq U.$$

By definition, this shows that U is in the topology generated by $\mathscr{B}$. Finally, we conclude that the topology generated by $\mathscr{B}$ is the standard topology.

Remark 1.2.14. Example 1.2.13 shows that there is a *countable* basis for the standard topology on $\mathbb{R}$. This will turn out to be a very useful fact.

Exercise 1.2.15. Let X be any nonempty set. Show that the collection of singletons (one-point subsets) of X is a basis for the discrete topology on X.

Example 1.2.16. In $\mathbb{R}$, consider

$$\mathscr{B} = \{[a, b) \mid a < b\}.$$

Note that the intersection of two elements of $\mathscr{B}$ is either empty or also an element of $\mathscr{B}$. Moreover, given $[a, b)$ and $[c, d)$ in $\mathscr{B}$, assume without loss of generality that $a \leqslant c$. If $x \in [a, b) \cap [c, d)$ then in particular $[a, b) \cap [c, d)$ is nonempty, and thus $c \leqslant x < b$. Hence $c < b$ and

$$x \in [x, b) \subseteq [c, b) \subseteq [a, b) \cap [c, d).$$

This shows that $\mathscr{B}$ is basis for a topology on $\mathbb{R}$, which is known as the **lower limit topology**. We will denote the topological space given by $\mathbb{R}$ with the lower limit topology by $\mathbb{R}_\ell$. A subset of $\mathbb{R}$ is open in $\mathbb{R}_\ell$ if and only if it is a union of half-open intervals of the form $[a, b)$.

Exercise 1.2.17. Show that the lower limit topology is strictly finer than the standard topology on $\mathbb{R}$.

Exercise 1.2.18. Let $\mathscr{B}_1$ and $\mathscr{B}_2$ be bases for a topology on X. Show that if $\mathscr{B}_1 \subseteq \mathscr{B}_2$, then the topology generated by $\mathscr{B}_2$ is finer than the topology generated by $\mathscr{B}_1$.

Exercise 1.2.19. Let $\mathscr{B}$ and $\mathscr{B}'$ be bases for topologies on the same set X, and let $\mathcal{T}$ and $\mathcal{T}'$ be the topologies generated by $\mathscr{B}$ and $\mathscr{B}'$ respectively. Show that the following are equivalent:

a) The topology $\mathcal{T}'$ is finer that $\mathcal{T}$.

b) For each $x \in X$ and $B \in \mathscr{B}$ with $x \in B$, there exists $B' \in \mathscr{B}'$ such that $x \in B' \subseteq B$.

Example 1.2.20. We can now compare the lower limit topology on $\mathbb{R}$ to the standard topology. Given any point $x \in \mathbb{R}$ and any basis element $B = (a, b)$ for the standard topology such that $x \in B$, note that we necessarily have $a < x < b$, so $B' = [x, b)$ is a basic open set for the lower limit topology and

$$x \in B' \subseteq B.$$

By Exercise 1.2.19, the lower limit topology $\mathbb{R}_\ell$ is finer than the standard topology on $\mathbb{R}$.

On the other hand, these two topologies do *not* coincide: for example, $[0, 1)$ is a basic open set in the lower limit topology with $0 \in [0, 1)$, but there are no open sets in the standard topology of $\mathbb{R}$ that contain 0 and are contained in $[0, 1)$. Thanks to Exercise 1.2.19, it even suffices to check that there are no open intervals (a, b) containing 0 and contained in $[0, 1)$.

We can now give an example of two incomparable topologies on the same set:

Exercise 1.2.21. Let

$$K := \left\{ \frac{1}{n} \mid n \in \mathbb{Z}, n \geqslant 1 \right\}.$$

Show that

$$\{(a, b) \mid a < b\} \cup \{(a, b) \setminus K \mid a < b\}$$

is a basis for a topology on $\mathbb{R}$, known as the **K-topology** on $\mathbb{R}$. The topological space given by $\mathbb{R}$ with the K-topology is denoted by $\mathbb{R}_K$.

Exercise 1.2.22. Show that the topological spaces $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are incomparable.

Example 1.2.23. Let $\mathscr{B}$

$$\mathscr{B} = \{(a, b) \times (c, d) \mid a < b, c < d\}$$

be the collection of open rectangles in $\mathbb{R}^2$. Any point in $\mathbb{R}^2$ belongs to some open rectangle, and the intersection of two open rectangles is an open rectangle, so this is a basis for a topology on $\mathbb{R}^2$, which we will temporarily call the rectangle topology. Let us compare it to the standard topology on $\mathbb{R}^2$.

We saw in Example 1.2.12 that the the set of open balls is a basis for the standard topology on $\mathbb{R}^2$. Given any point x in $\mathbb{R}^2$ and any rectangle L containing it, any open ball $B = B_\varepsilon(x)$ centered at x with radius ε small enough will be contained in L. Thus by Exercise 1.2.19, the standard topology is finer than the rectangle topology. Conversely, given any point x and any open ball containing it, we can find an open rectangle containing x that is small enough to be contained in the ball.[1] We conclude that in fact the two topologies are the same, and hence we will no longer refer to it as the rectangle topology.

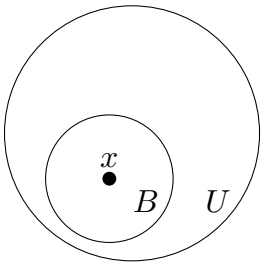
[1] Think about how could make this argument more precise. Perhaps start by taking the smallest distance of x to a point in the boundary circle of the ball, and think of a precise rectangle we could define that contains x and is contained in our circle.

This example illustrates a useful way to check that a collection of open sets is actually a basis for a given topology:

Lemma 1.2.24. *Let $(X, \mathcal{T})$ be a topological space. Suppose $\mathscr{B}$ is a collection of open subsets of X (meaning that $\mathscr{B} \subseteq \mathcal{T}$) such that the following condition holds:*

for every open set U and every $x \in U$, there exists $B \in \mathscr{B}$ such that

$$x \in B \subseteq U.$$



Then $\mathscr{B}$ is a basis for the topology $\mathcal{T}$.

Proof. We have to check two things: that $\mathscr{B}$ is a basis and that the topology generated by $\mathscr{B}$ coincides with $\mathcal{T}$. First, we prove that $\mathscr{B}$ is in fact a basis for a topology. Given $x \in X$, since X is open, by assumption there exists $B \in \mathscr{B}$ such that $x \in B \subseteq X$. In particular,

$$\bigcup_{B \in \mathscr{B}} B = X.$$

Now given $B_1, B_2 \in \mathscr{B}$ and $x \in B_1 \cap B_2$, note that B_1 and B_2 are by assumption open sets, so $B_1 \cap B_2$ is open, and thus by assumption there exists $B_3 \in \mathscr{B}$ such that $x \in B_3 \subseteq B_1 \cap B_2$. Thus $\mathscr{B}$ is a basis for a topology on X. Let $\mathcal{T}'$ be the topology generated by $\mathscr{B}$.

Now note that $\mathcal{T}$ is a basis for itself. Since $\mathscr{B}$ is a basis for $\mathcal{T}'$, and $\mathscr{B} \subseteq \mathcal{T}$, Exercise 1.2.18 applies and implies that $\mathcal{T}' \subseteq \mathcal{T}$.

Let U be any open set in $\mathcal{T}$. By assumption, for each $x \in U$ there exists $B_x \in \mathscr{B}$ such that $x \in B_x \subseteq U$. In particular,

$$U = \bigcup_{x \in U} \{x\} \subseteq \bigcup_{x \in U} B_x \subseteq U,$$

so U is a union of basic open sets in $\mathscr{B}$. By Theorem 1.2.9, U is an open set in $\mathcal{T}'$, the topology generated by $\mathscr{B}$. We conclude that $\mathcal{T} \subseteq \mathcal{T}'$ and thus the two topologies coincide. □

Remark 1.2.25. We can now rewrite Lemma 1.2.24 in a nice compact way: a collection of open sets $\mathscr{B}$ is a basis for the topology on X if and only if for all $x \in X$, every neighborhood of x contains a neighborhood of X contained in $\mathscr{B}$.

Definition 1.2.26. A **subbasis** for a topology on a set X is a collection $S \subseteq \mathcal{P}(X)$ of subsets of X such that

$$\bigcup_{U \in S} U = X.$$

Exercise 1.2.27. Let S be a subbasis for a topology on S. Show that the collection $\mathscr{B}$ of all finite intersections of elements of S is a basis for a topology on X. The topology generated by $\mathscr{B}$ is the **topology generated by** S.

Remark 1.2.28. The topology generated by a subbasis S is the smallest topology containing S: any topology containing S must contain the basis $\mathscr{B}$ described in Exercise 1.2.27, and the topology generated by $\mathscr{B}$.

1.3 Closed sets

Definition 1.3.1. Let $(X, \mathcal{T}_X)$ be a topological space. A subset $C \subseteq X$ is **closed** if its complement $X \setminus C$ is open.

Remark 1.3.2. Unlike doors, subsets need not be either open or closed. For instance, in $\mathbb{R}$ with the standard topology, the subset $[0, 1)$ is neither open nor closed. Conversely, a set can be both open and closed at the same time; such a set is called **clopen**. In $X = \{a, b\}$ with the discrete topology, $\{a\}$ is clopen.

Remark 1.3.3. Let X be a topological space. Using basic set theory, we can take the definition of open sets to given the following equivalent characterization of closed sets:

- The empty set is closed,
- The entire space X is closed.
- The *union* of two closed sets is closed, and
- The arbitrary *intersection* of closed sets is closed.

Remark 1.3.4. In fact, the axioms of a topology could just as well have been given in terms of closed subsets: a topology on X is given by a collection of subsets $\mathcal{C}$, known as the closed subsets of X, satisfying the four axioms in Remark 1.3.3. We would then define open sets as those whose complements are closed.

Lemma 1.3.5. *Let X be a topological space. A subset $C \subseteq X$ is closed if and only if for all $x \notin C$ there exists an open set U such that*

$$x \in U \quad \text{and} \quad U \cap C = \emptyset.$$

Moreover, given a basis for the topology on X, such U can be taken to be a basic open set.

Proof. ($\Rightarrow$) If C is closed, then for all $x \notin C$, since $X \setminus C$ is open, we can find an open set U (which we can even take to be a basic open set, by Remark 1.2.25) such that $x \in U \subseteq X \setminus C$, or equivalently, $x \in U$ and $U \cap C = \emptyset$.

($\Leftarrow$) Suppose that for all $x \notin C$ there exists an open set U such that $x \in U$ and $U \cap C = \emptyset$. Then $x \in U \subseteq X \setminus C$, and thus $X \setminus U$ is open by Exercise 1.1.20. □

Remark 1.3.6. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be topologies on a set X, and assume that for all closed sets C for the topology $\mathcal{T}_1$, C is also closed in $\mathcal{T}_2$. Then given any open set $U \in \mathcal{T}_1$, $X \setminus U$ is closed in $\mathcal{T}_1$, and thus closed in $\mathcal{T}_2$, so U must be open in $\mathcal{T}_2$. Thus we can check that $\mathcal{T}_1 \subseteq \mathcal{T}_2$ by checking that every closed set in $\mathcal{T}_1$ is closed in $\mathcal{T}_2$.

Example 1.3.7. Let X be any set equipped with the finite complement topology. Then the closed subsets are X and all the finite subsets (including the empty set).

Example 1.3.8. In $\mathbb{R}$ with the standard topology, closed intervals are closed. These are any sets of the form $[a, b]$ with $a < b$, since

$$[a, b] = \mathbb{R} \setminus ((-\infty, a) \cup (b, \infty)).$$

In fact, note that any singletons $\{a\}$ are also closed: given any real number $x \neq a$, we can find a small enough open neighborhood U_x of x such that $a \notin U_x$, and thus

$$\bigcup_{x \neq a} U_x = \mathbb{R} \setminus \{a\}$$

is open. If we want to make this precise, we can take $\varepsilon > 0$ with $\varepsilon < |x - a|$, and take $U_x = B_\varepsilon(x)$ to be the ball of radius ε centered at x.

More generally, finite unions of closed intervals are closed, and arbitrary intersections of closed intervals are closed. In particular, any finite set in $\mathbb{R}$ is a closed set, since it is the union of finitely many singletons, which are closed.

Example 1.3.9 (the Cantor set)**.** Consider following sequence of subsets of $\mathbb{R}$:

$$\begin{aligned} C_0 &= [0, 1] \\ C_1 &= [0, 1] \setminus (1/3, 2/3) = [0, 1/3] \cup [2/3, 1] \\ C_2 &= C_1 \setminus (1/9, 2/9) \cup (7/9, 8/9) = [0, 1/9] \cup [2/9, 1/3] \cup [2/3, 7/9] \cup [8/9, 1] \end{aligned}$$

and so on, so that for each $n \geqslant 1$

$$C_n = \frac{C_{n-1}}{3} \cup \left(\frac{2}{3} + \frac{C_{n-1}}{3} \right)$$

where scaling and adding sets of real numbers is defined in the obvious way (by adding or scaling elements). Note that each C_n is closed in $\mathbb{R}$ for the standard topology, since it is a finite union of closed intervals. It follows that the **Cantor set**

$$C := \bigcap_{n \geqslant 0} C_n$$

is also closed. One way to describe C is that it consists of those real numbers that admit a base three expansion of the form

$$0.i_1 i_2 i_3 \cdots_3 \quad \text{with } i_j \in \{0, 2\} \quad \text{for all } j.$$

For example, $1/3 = 0.1_3$ is in the Cantor set since it can also be written as $1/3 = 0.0\overline{2}_3$.

In fact, note that an element in

$$\bigcap_{n=0}^{m} C_n$$

for some $m \geqslant 1$ is a real number whose base three expansion $0.i_1 i_2 i_3 \cdots_3$ satisfies $i_j \in \{0, 2\}$ for all $j \leqslant m$.

Example 1.3.10. Consider $\mathbb{R}_\ell$, the real numbers with the lower limit topology. The set $(-\infty, 0)$ is closed, since its complement

$$\mathbb{R} \setminus (-\infty, 0) = [0, \infty) = \bigcup_{r > 0} [0, r)$$

is a union of open sets, and thus open.

1.4 The product topology for finite products

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. Our goal is to equip the cartesian product

$$X \times Y = \{(x, y) \mid x \in X, y \in Y\}$$

with a reasonable topology.

Theorem 1.4.1. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces. The set*

$$\mathscr{B} := \{U \times V \mid U \in \mathcal{T}_X \text{ and } V \in \mathcal{T}_Y\}$$

is a basis for a topology on $X \times Y$, called the **product topology**. *More generally, if $\mathscr{B}_X$ is a basis for $\mathcal{T}_X$ and $\mathscr{B}_Y$ is a basis for $\mathcal{T}_Y$, then*

$$\mathscr{B}_{X\times Y} := \{U \times V \mid U \in \mathscr{B}_X \text{ and } V \in \mathscr{B}_Y\}$$

is also a basis for the product topology on $X \times Y$.

Proof. Let $(x, y) \in X \times Y$. By definition of basis, there exist $U \in \mathscr{B}_X$ and $V \in \mathscr{B}_Y$ such that $x \in U$ and $y \in V$. Therefore, $(x, y) \in U \times V$ with $U \times V \in \mathscr{B}_{X\times Y}$. This proves the first axiom for a basis is satisfied.

Suppose $(x, y) \in U \times V \cap U' \times V'$, where $U, U' \in \mathscr{B}_X$ and $V, V' \in \mathscr{B}_Y$. Then $x \in U \cap U'$ and $y \in V \cap V'$. Since $\mathscr{B}_X$ and $\mathscr{B}_Y$ are bases for $\mathcal{T}_X$ and $\mathcal{T}_Y$, respectively, there exists $U'' \in \mathscr{B}_X$ such that

$$x \in U'' \subseteq U \cap U'.$$

Analogously, there exists $V'' \in \mathscr{B}_Y$ with

$$y \in V'' \subseteq V \cap V'.$$

Therefore,

$$(x, y) \in U'' \times V'' \subseteq U \times V \cap U' \times V'.$$

This concludes the proof that $\mathscr{B}_{X\times Y}$ is a basis for a topology on $X \times Y$, proving the second part of our statement. It remains to show that $\mathscr{B}$ and $\mathscr{B}_{X\times Y}$ are basis for the same topology.

Now note that we can take $\mathscr{B}_X = \mathcal{T}_X$ and $\mathscr{B}_Y = \mathcal{T}_Y$ as bases for each of the topologies on X and Y, respectively, which proves $\mathscr{B}$ is also a basis for a topology on $X \times Y$. Note that

$$\mathscr{B}_{X\times Y} \subseteq \mathscr{B},$$

so by Exercise 1.2.18 the topology generated by $\mathscr{B}$ is finer than the product topology.

Conversely, fix $(x, y) \in X \times Y$ and $B \in \mathscr{B}$, and say $B = U \times V$ for some $U \in \mathcal{T}_X$ and $V \in \mathcal{T}_Y$. Since $\mathscr{B}_X$ and $\mathscr{B}_Y$ are bases for the respective topologies, we can find $B_x \in \mathscr{B}_X$ and $B_y \in \mathscr{B}_Y$ such that such that

$$x \in B_x \subseteq U \quad \text{and} \quad y \in B_y \subseteq V.$$

Then

$$(x, y) \in B_x \times B_y \subseteq U \times V.$$

By Exercise 1.2.19, the product topology is finer than the topology generated by $\mathscr{B}$. We conclude that the two bases generated the same topology. □

Remark 1.4.2. In general, given topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$, the set

$$\{U \times V \mid U \in \mathcal{T}_X \text{ and } V \in \mathcal{T}_Y\}$$

is not a topology on $X \times Y$, and we truly need to consider unions of such sets.

Example 1.4.3. We saw in Example 1.2.10 that the set of open intervals forms a basis for the standard topology on $\mathbb{R}$, so the set of open rectangles forms a basis for the product topology on $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$. On the other hand, by Example 1.2.23 this basis generates the standard topology on $\mathbb{R}^2$. In summary: the product topology on $\mathbb{R}^2$ agrees with the standard topology.

This example illustrates Remark 1.4.2 perfectly: there are open sets in $\mathbb{R}^2$ that are not open rectangles.

Exercise 1.4.4. Consider $X = \mathbb{R}$ with the standard topology and $Y = \{0, 1\}$ with the discrete topology. Describe the product topology on $X \times Y$, by describing what any arbitrary open set looks like.

Remark 1.4.5. One can of course iterate this process. Given topological spaces $(X, \mathcal{T}_X)$, $(Y, \mathcal{T}_Y)$, and $(Z, \mathcal{T}_Z)$, we may consider $(X \times Y) \times Z$ and $X \times (Y \times Z)$, and take the product topologies on each. Note that as sets, these are technically not the same set: one has elements of the form $(x, (y, z))$ and the other elements of the form $((x, y), z)$. But there is of course a bijection between the two and $X \times Y \times Z$, whose elements are triples of the form (x, y, z) with $x \in X$, $y \in Y$, and $z \in Z$. We then consider the natural product topology, with basis

$$\{U \times V \times W \mid U \in \mathscr{B}_X, V \in \mathscr{B}_Y, W \in \mathscr{B}_Z\}$$

given any bases $\mathscr{B}_X$, $\mathscr{B}_Y$, and $\mathscr{B}_Z$ for the respective topological spaces. One can check that in fact this matches the definition of the product topology on $(X \times Y) \times Z$ and $X \times (Y \times Z)$.

Definition 1.4.6. Given topological spaces $(X_1, \mathcal{T}_1), \ldots, (X_n, \mathcal{T}_n)$, the **product topology** on $X_1 \times \cdots \times X_n$ is the topology generated by the basis

$$\{U_1 \times \cdots \times U_n \mid U_i \in \mathcal{T}_i\}.$$

One can also define a product topology on an arbitrary product of spaces. However, that carries with it some interesting new challenges, and it turns out there is more than one good choice of how to naturally define a topology on an arbitrary product of spaces. We will discuss this in more detail later.

Exercise 1.4.7. Let $\mathbb{R}^n$ denote $\overbrace{\mathbb{R} \times \cdots \times \mathbb{R}}^{n}$. Show that the product topology on

$$\mathbb{R}^n = \overbrace{\mathbb{R} \times \cdots \times \mathbb{R}}^{n},$$

where each factor $\mathbb{R}$ is given the standard topology, coincides with the standard topology in $\mathbb{R}^n$.

Exercise 1.4.8. Consider the space $\mathbb{R}_\ell$ consisting of $\mathbb{R}$ equipped with the lower limit topology, and let $\mathbb{R}$ be the real line equipped with the standard topology. Describe bases for $\mathbb{R}_\ell \times \mathbb{R}$ and $\mathbb{R}_\ell \times R_\ell$.

1.5 The order topology

Definition 1.5.1. Let X be a set. A **strict partial order** is a binary relation $<$ on X satisfying the following properties:

- (Irreflexivity) For all $x \in X$, we have $x \not< x$, meaning $x < x$ does not hold.
- (Transitivity) If $x < y$ and $y < z$, then $x < z$.

A **strict total order** is a strict partial order satisfying the following additional property:

- (Total) If $x \neq y$, then $x < y$ or $y < x$.

We write $x \leqslant y$ to denote that $x < y$ or $x = y$.

Notation 1.5.2. Let X be a set with a strict total order $<$. Given elements $a < b$ in X, the **intervals** determined by a and b are the following subsets of X:

- The **open interval**
$$(a, b) := \{x \in X \mid a < x < b\}.$$
- The **closed interval**
$$[a, b] := \{x \in X \mid a \leqslant x \leqslant b\}.$$
- The **half open intervals**
$$(a, b] := \{x \in X \mid a < x \leqslant b\} \quad \text{and} \quad [a, b) := \{x \in X \mid a \leqslant x < b\}.$$

Remark 1.5.3. One may also use the alternative notation for open intervals
$$]a, b[:= \{x \in X \mid a < x < b\}.$$
This comes in handy when dealing with spaces that are themselves products, say $X \times Y$, where the points are pairs (x, y), as it can become confusing to *also* use the notation (a, b) for open intervals.

Definition 1.5.4. Let X be a set with a strict total order $<$. We let $\mathscr{B}_<$ denote the collection of all of the following subsets of X:

- For all $a < b$, we let $(a, b) \in \mathscr{B}_<$.
- If there exists $m \in X$ such that $m \leqslant x$ for all $x \in X$, we let $[m, b) \in \mathscr{B}_<$ for all $b \in X$ with $m < b$.
- If there exists $M \in X$ such that $x \leqslant M$ for all $x \in X$, we let $(a, M] \in \mathscr{B}_<$ for all $a < M$.

Show that $\mathscr{B}_<$ is a basis for a topology on X.

Definition 1.5.5. Let X be a set with a strict total order $<$. The **order topology** induced by $<$ on X is the topology generated by $\mathscr{B}_<$ as in Definition 1.5.4.

Example 1.5.6. The order topology on $\mathbb{R}$ is the standard topology.

Example 1.5.7. The order topology on $\mathbb{N}$ is the discrete topology: indeed, all points are open, since $\{0\} = [0, 1)$ and $\{n\} = (n-1, n+1)$ for $n \geqslant 1$.

Definition 1.5.8. Let $(X, <_X)$ and $(Y, <_Y)$ be two sets with a strict total order. The **dictionary order** on $X \times Y$ is the strict total order defined by

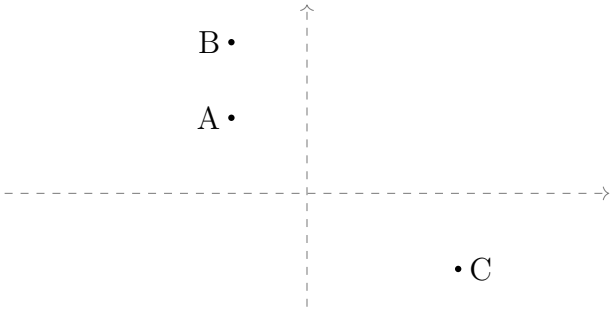
$$(x, y) < (a, b) \text{ if } x < a \text{ or } x = a \text{ and } y < b.$$

The **dictionary topology** on $X \times Y$ is the topology induced by the dictionary order.

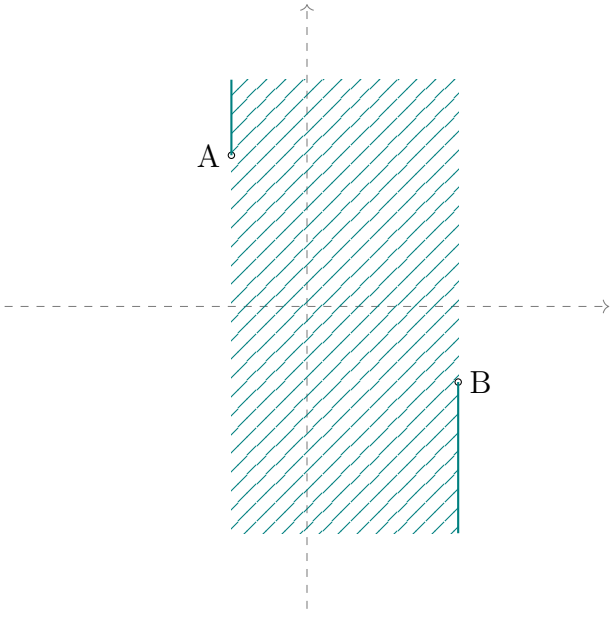
Example 1.5.9. We can describe the dictionary order on $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$ informally as follows: given A and B in $\mathbb{R}^2$, $A < B$ if one of the following holds:

- A and B are on the same vertical line and A is below B, or
- A is to the left of B.

For example, in the picture below, $A < B < C$:



The picture below shows the interval (A, B), which is a basic open set in $\mathbb{R}^2$ with the dictionary topology. Note that A and B are *not* included in the interval.



Exercise 1.5.10. Consider the usual order on $\mathbb{N}$. Show that the order topology on $\mathbb{N} \times \mathbb{N}$ does not agree with the product topology: it is not the discrete topology.

1.6 The subspace topology

Theorem 1.6.1. *Let $(X, \mathcal{T}_X)$ be a topological space and Y any subset of X. Set*

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}_X\}.$$

Then $\mathcal{T}_Y$ is a topology on Y.

Proof. Since $\emptyset \in \mathcal{T}_X$ and $X \in \mathcal{T}_X$,

$$\emptyset = \emptyset \cap Y \in \mathcal{T}_Y \quad \text{and} \quad Y = X \cap Y \in \mathcal{T}_Y.$$

Fix U and V in $\mathcal{T}_Y$, and let $A, B \in \mathcal{T}_X$ be such that $U = A \cap Y$ and $V = B \cap Y$. Since $\mathcal{T}_X$ is a topology, $A \cap B \in \mathcal{T}_X$. Therefore,

$$U \cap V = (A \cap Y) \cap (B \cap Y) = \underbrace{(A \cap B)}_{\in \mathcal{T}_X} \cap Y.$$

Finally, given a collection $\{U_\alpha\}$ of elements of $\mathcal{T}_Y$ indexed by a set Λ, where each U_α is of the form $U_\alpha = V_\alpha \cap Y$ for some open set $V_\alpha \in \mathcal{T}_X$ in X, since $\mathcal{T}_X$ is a topology we have

$$\bigcup_{\alpha \in \Lambda} V_\alpha \in \mathcal{T}_X.$$

Thus

$$\bigcup_{\alpha \in \Lambda} U_\alpha = \bigcup_{\alpha \in \Lambda} (V_\alpha \cap Y) = \underbrace{\left(\bigcup_{\alpha \in \Lambda} V_\alpha \right)}_{\in \mathcal{T}_X} \cap Y \in \mathcal{T}_Y.$$

This finishes the proof that $\mathcal{T}_Y$ is a topology. □

Definition 1.6.2. Given a topological space $(X, \mathcal{T}_X)$, the **subspace topology** on $Y \subseteq X$ is the topology with open sets

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}_X\}.$$

We may say Y is a subspace of X to indicate that X is a topological space and that we consider $Y \subseteq X$ with the subspace topology.

Exercise 1.6.3. Identify $\mathbb{R}$ with $\mathbb{R} \times \{0\} \subseteq \mathbb{R}^2$ in the natural way. Show that the subspace topology on $\mathbb{R}$ coincides with the standard topology.

Example 1.6.4. Consider the standard topology on $\mathbb{R}^2$. The topological space S^1 is the subset of $\mathbb{R}^2$ given by

$$S^1 := \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$

equipped with the subspace topology.

More generally, consider the standard topology on $\mathbb{R}^{n+1}$. The **sphere** S^n is given by

$$S^n := \{(x_1, \ldots, x_{n+1}) \in \mathbb{R}^{n+1} \mid x_1^2 + \cdots + x_{n+1}^2 = 1\}$$

equipped with the subspace topology. The n-dimensional **disk** D^n is

$$D^n := \{(x_1, \ldots, x_n) \in \mathbb{R}^n \mid x_1^2 + \cdots + x_n^2 \leqslant 1\}$$

equipped with the subspace topology.

Lemma 1.6.5. *Let Y be a subspace of X. A subset $A \subseteq Y$ is closed in Y if and only if $A = C \cap Y$ for some closed set C of X.*

Proof. ($\Leftarrow$) If C is a closed set of X, then $X \setminus C$ is open in X, and

$$Y \setminus C = (X \setminus C) \cap Y$$

is open in Y. Therefore, C is closed in Y.

($\Rightarrow$) Suppose C is closed in Y, so that $Y \setminus C$ is open in Y. Then there exists an open set U of X such that

$$Y \setminus C = U \cap Y.$$

Then $X \setminus U$ is closed in X, and

$$C = Y \setminus (U \cap Y) = Y \setminus U = (X \setminus U) \cap Y$$

is open in Y. □

Remark 1.6.6. Let X be a topological space and $Z \subseteq Y \subseteq X$. If Z is open in X, then it is also open in Y with the subspace topology, as $Z = Z \cap Y$. Similarly, if Z is closed in X, then Z is also closed in Y with the subspace topology.

Remark 1.6.7. Suppose Y itself is an open subset of X, equipped with the subspace topology. Then given any open subset U of X, $U \cap Y$ is open in Y (by definition), but also open in X since it is an intersection of open sets (of X). Thus whenever Y itself is open in X, the open subsets of Y for the subspace topology coincide with those open subsets of X that happen to be contained in Y. Analogously, if Y is a closed subset of a topological space X and we endow Y with the subspace topology, then the closed subsets of Y are those subsets of Y that are closed in X.

However, the converse does not hold.

Example 1.6.8. Consider $\mathbb{R}$ with the standard topology, and endow $Y = [0, 1]$ with the subspace topology. Note that some of the open sets in $[0, 1]$ do not remain open when we consider them as subsets of $\mathbb{R}$: for example, $(0, 1] = (0, 2) \cap Y$ is open in Y but not in $\mathbb{R}$.

Example 1.6.9. Consider $\mathbb{R}$ with the standard topology, and endow $Y = (0, 1)$ with the subspace topology. Then $(0, \frac{1}{2}] = [-1, \frac{1}{2}] \cap Y$ is closed in Y but not in $\mathbb{R}$.

Remark 1.6.10. Example 1.6.8 shows that when Y is a subspace of a topological space X and we want to talk about a subset U of Y, there is an important distinction between *open in Y* and *open in X*, and thus we should always avoid simply saying *open* without indicating *in what space*. The analogous statement applies to closed subsets, as Example 1.6.9 shows.

Before we give more examples, the following unsurprising lemma will be helpful:

Lemma 1.6.11. *Let $\mathscr{B}_X$ be a basis for the topology on a topological space $(X, \mathcal{T}_X)$, and Y is any subset of X. Then*

$$\mathscr{B}_Y := \{U \cap Y \mid U \in \mathscr{B}_X\}$$

is a basis for the subspace topology on Y.

Proof. First, note that every element in $\mathscr{B}_Y$ is an open set in the subspace topology of Y, so we can apply Lemma 1.2.24 to check that $\mathscr{B}_Y$ is indeed a basis for the subspace topology.

Any open subset of Y has the form $U \cap Y$ for some open set U in $\mathcal{T}_X$. Let $y \in U \cap Y$. Since U is open in X and $\mathscr{B}_X$ is a basis for $\mathcal{T}_X$, there exists $B \in \mathscr{B}_X$ such that $x \in B \subseteq U$. Therefore, $y \in B \cap Y \subseteq U \cap Y$. By Lemma 1.2.24, $\mathscr{B}_Y$ is a basis for the subspace topology. □

Example 1.6.12. Consider $\mathbb{R}$ with the standard topology, and endow $[0,1]$ with the subspace topology. Since the collection of open intervals is a basis for the topology on $\mathbb{R}$, by Lemma 1.6.11 we can obtain a basis for the subspace topology on $[0,1]$ by intersecting all open intervals on $\mathbb{R}$ with $[0,1]$. We conclude that

$$\{(a,b) \mid 0 < a < b < 1\} \cup \{[0,b) \mid 0 < b < 1\} \cup \{(a,1] \mid 0 < a < 1\} \cup \{\emptyset\}$$

is a basis for the subspace topology on $[0,1]$.

Lemma 1.6.13. *Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be topological spaces, and $A \subseteq X$ and $B \subseteq Y$ be subsets equipped with the corresponding subspace topology. The product topology on $A \times B$ coincides with the subspace topology on $A \times B \subseteq X \times Y$ when $X \times Y$ is given the product topology.*

Proof. When regarding $A \times B$ with the product topology, we have a basis given by

$$\mathscr{B}_{\text{prod}} = \{(U \cap A) \times (V \cap B) \mid U \in \mathcal{T}_X \text{ and } V \in \mathcal{T}_Y\}.$$

When regarding $A \times B$ under the subspace topology, combining Lemma 1.6.11 and Theorem 1.4.1 gives us the basis

$$\mathscr{B}_{\text{sub}} = \{(U \times V) \cap (A \times B) \mid U \in \mathcal{T}_X \text{ and } V \in \mathcal{T}_Y\}.$$

Basic properties of set theory (see Exercise A.3.5) tell us that

$$(U \cap A) \times (V \cap B) = (U \times V) \cap (A \times B).$$

We conclude that the two bases are actually equal, and thus the two topologies coincide. □

1.7 Closure and interior

Definition 1.7.1. Let $(X, \mathcal{T}_X)$ be a topological space and let $S \subseteq X$ be any subset. The **closure** of S (in X), written as $\overline{S}^X$ or simply $\overline{S}$ if X is clear from context, is the intersection of all the closed subsets of X that contain S:

$$\overline{S} := \bigcap_{\substack{S \subseteq C \\ C \text{ is closed in } X}} C.$$

Remark 1.7.2. Since an arbitrary intersection of closed subsets is closed, the closure $\overline{S}$ of any subset S is closed. In fact, $\overline{S}$ is the unique smallest closed subset of X that contains S. In particular, S is closed if and only if $S = \overline{S}$.

Definition 1.7.3. Let $(X, \mathcal{T}_X)$ be a topological space and let $S \subseteq X$ be any subset. The **interior** of S, written S^{int}, is the union of all open subsets of X contained in S:

$$S^{\text{int}} := \bigcup_{\substack{U \subseteq S \\ U \text{ is open in } X}} U.$$

Remark 1.7.4. Since an arbitrary union of open subsets is open, the interior S^{int} of any subset S is open. In fact, it is the unique largest open set contained in S. In particular, S is open if and only if $S = S^{\text{int}}$.

Remark 1.7.5. Note that for any subset A of a topological space X, $A^{\text{int}} \subseteq A \subseteq \overline{A}$.

Exercise 1.7.6. Show that if $A \subseteq B$ then $\overline{A} \subseteq \overline{B}$ and $A^{\text{int}} \subseteq B^{\text{int}}$.

Theorem 1.7.7. *Let $(X, \mathcal{T}_X)$ be a topological space and let $S \subseteq X$ be any subset. Fix a basis $\mathscr{B}$ for the topology on X.*

a) Given $x \in X$,

$$\begin{aligned} x \in S^{int} &\iff \textit{there exists an open neighborhood } U \textit{ of } x \textit{ such that } U \subseteq S \\ &\iff \textit{there exists a basic open neighborhood } U \in \mathscr{B} \textit{ of } x \textit{ such that } U \subseteq S. \end{aligned}$$

b) Given $x \in X$,

$$\begin{aligned} x \in \overline{S} &\iff \textit{every open neighborhood } U \textit{ of } x \textit{ has } U \cap S \neq \emptyset \\ &\iff \textit{every basic open neighborhood } U \in \mathscr{B} \textit{ of } x \textit{ has } U \cap S \neq \emptyset. \end{aligned}$$

Proof. By definition,

$$S^{\text{int}} := \bigcup_{\substack{U \subseteq S \\ U \text{ is open in } X}} U.$$

An element x is in this union if and only if x is one of the sets in the union, meaning that $x \in U$ for some open set $U \subseteq S$. Moreover, any open set is a union of basic opens, so $x \in U$ for some open set $U \subseteq S$ if and only if $x \in U$ for some basic open $U \subseteq S$.

To show the second statement, we prove the contrapositive of each implication.

($\Rightarrow$) Suppose there is an open neighborhood U of x with $U \cap S = \emptyset$. Then $S \subseteq X \setminus U$, and $X \setminus U$ is closed by definition. By definition of closure, $\overline{S} \subseteq X \setminus U$. On the other hand, $x \notin X \setminus U$, so $x \notin \overline{S}$.

($\Leftarrow$) Let

$$x \notin \overline{S} = \bigcap_{\substack{S \subseteq C \\ C \text{ is closed in } X}} C.$$

Thus $x \notin C$ for some closed subset C of X with $S \subseteq C$. Then $X \setminus C$ is an open neighborhood of x that does not intersect S. Since $X \setminus C$ is a union of basic open sets, we can find $U \in \mathscr{B}$ with $U \subseteq X \setminus C$, and U does not intersect S since it does not intersect $C \supseteq S$. □

Exercise 1.7.8. Consider the subset $S = (a, b)$ of $\mathbb{R}$.

a) Find the closure of S when $\mathbb{R}$ is given the standard topology.

b) Find the closure of S when $\mathbb{R}$ is given the lower limit topology.

c) Find the interior of S when $\mathbb{R}$ is given the standard topology.

d) Find the interior of S when $\mathbb{R}$ is given the lower limit topology.

The complement of the closure is the interior of the complement:

Lemma 1.7.9. *Let S be a subset of a topological space $(X, \mathcal{T})$. Then*

$$X \setminus \overline{S} = (X \setminus S)^{int}.$$

Proof. Fix $x \in X$. Then

$$\begin{aligned} x \in X \setminus \overline{S} &\iff x \notin \overline{S} \\ &\iff \text{there is an open neighborhood } U \text{ of } x \text{ with } U \cap S = \emptyset \quad \text{by Theorem 1.7.7} \\ &\iff \text{there is an open neighborhood } U \text{ of } x \text{ with } U \subseteq X \setminus S \\ &\iff x \in (X \setminus S)^{\text{int}} \quad \text{by Theorem 1.7.7.} \end{aligned}$$

□

Definition 1.7.10. A subset A of a topological space X is **dense** in X if $\overline{A} = X$.

Example 1.7.11. Let us find the closure of $\mathbb{Q}$ in $\mathbb{R}$ under the standard topology. Given $x \in \mathbb{R}$ and any open neighborhood U of x in $\mathbb{R}$, $(x - \varepsilon, x + \varepsilon) \subseteq U$ for some $\varepsilon > 0$. By the Density of the Rationals in the real numbers, there exists $q \in \mathbb{Q}$ with $x < q < x + \varepsilon$, and hence

$$U \cap \mathbb{Q} \neq \emptyset.$$

Therefore, $x \in \overline{\mathbb{Q}}$, and we conclude that $\overline{\mathbb{Q}} = \mathbb{R}$. Therefore, $\mathbb{Q}$ is dense in $\mathbb{R}$.

Exercise 1.7.12. Consider $X = [0, 1]$ with the subspace topology for the standard topology on $\mathbb{R}$. Let C be the Cantor set and $U = X \setminus C$. Find the closure of U in X.

Exercise 1.7.13. Consider $\mathbb{R}$ with the standard topology and the subset $K = \{\frac{1}{n} \mid n \in \mathbb{N}\}$.

a) Find the closure of K in $\mathbb{R}$ with the standard topology.

b) Find the closure of K in $\mathbb{Q}$ with the subspace topology.

Exercise 1.7.14. Let Y be a subspace of X and S is any subset of Y. Show that

$$\overline{S}^Y = \overline{S}^X \cap Y.$$

Definition 1.7.15. For a subset S of a topological space X, an **accumulation point** or **limit point** of S (in X) is a point $x \in X$ such that every open neighborhood of x (in X) contains a point of S other than x itself. We write S' for the set of all accumulation points of S.

Remark 1.7.16. Note that by Theorem 1.7.7, x is an accumulation point of S if and only if $x \in \overline{S \setminus \{x\}}$.

Lemma 1.7.17. *Let S be any subset of a topological space X, and let S' be the set of accumulation points of S. Then*

$$\overline{S} = S \cup S'.$$

Proof. ($\supseteq$) If $x \in S'$, then in particular every open neighborhood of x intersects S, so $x \in \overline{S}$ by Theorem 1.7.7. Moreover, $S \subseteq \overline{S}$ by definition.

($\subseteq$) Let $x \in \overline{S}$. If $x \in S$, there is nothing to show. Suppose $x \notin S$ and let U be an open neighborhood of x. Since $x \in \overline{S}$, then U intersects S nontrivially by Theorem 1.7.7, and since $x \notin S$, $U \cap S$ must contain a point other than x. We conclude that $x \in S'$. □

Corollary 1.7.18. *A subset S of a topological space X is closed if and only if S contains all its accumulation points.*

But, not every point in S is necessarily an accumulation point:

Exercise 1.7.19. Let $K = \{\frac{1}{n} \mid n \in \mathbb{N}\}$, a subset of $\mathbb{R}$ with the standard topology. Show that the only accumulation point of S is 0. In particular, $S' \cap S = \emptyset$.

Example 1.7.20. The fact that the rationals are dense in $\mathbb{R}$ gives us $\mathbb{Q}' = \mathbb{R}$.

Definition 1.7.21. A sequence $(x_n)_{n \geqslant 1}$ of points in topological space X **converges to a point** $y \in X$ if for all open neighborhoods U of y there exists an integer $N \geqslant 1$ such that $x_n \in U$ for all $n \geqslant N$.

A strange thing about converging sequences in a general topological space is that a sequence can converge to many points, not necessarily just one.

Example 1.7.22. Consider any set X with the indiscrete topology. Any sequence of points in X converges to all points in X: given any point $x \in X$, its unique open neighborhood is X, and any sequence of points in X is completely contained in that neighborhood.

Exercise 1.7.23. Let y and x_n for all $n \geqslant 1$ be real numbers.

a) In $\mathbb{R}$ in the standard topology, prove that (x_n) converges to y if and only if for all $\varepsilon > 0$ there exists $N \geqslant 1$ such that $|x_n - y| < \varepsilon$ for all $n \geqslant N$.

b) In $\mathbb{R}_\ell$, prove that (x_n) converges to y if and only if for all $\varepsilon > 0$ there exists N such that $y \leqslant x_n < y + \varepsilon$ for all $n \geqslant N$.

c) In $\mathbb{R}_{\text{discrete}}$, prove that (x_n) converges to y if and only if there exists N such that $y = x_n$ for all $n \geqslant N$.

Exercise 1.7.24. In the finite complement topology on $\mathbb{R}$, to what points (if any) does the sequence $\left(\frac{1}{n}\right)$ converge?

Exercise 1.7.25. Consider $\mathbb{R}$ with the included point topology with included point 0.

a) To what points (if any) does the constant sequence equal to 0 converge to?

b) To what points (if any) does the constant sequence equal to 1 converge to?

Remark 1.7.26. Let X be a topological space and $S \subseteq X$. If (x_n) is a sequence of points with values in S, and (x_n) converges to s, then every open neighborhood U of s must contain some x_n (in fact, all for n large), and thus $U \cap S \neq \emptyset$. We conclude that $s \in \overline{S}$. In summary, all the limits of a sequence with values in S must be in $\overline{S}$.

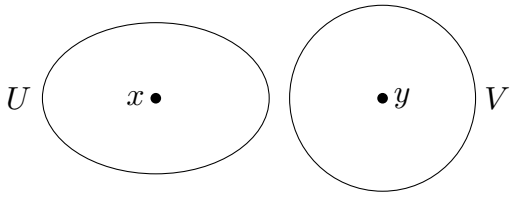
However, the converse does not always hold: in a general topological space, it need not be that all points in $\overline{S}$ are the limit of some sequence of points in S.

Definition 1.7.27. A topological space X is **Fréchet-Urysohn** if for every subset $S \subseteq X$ and every $s \in \overline{S}$, there exists a sequence (s_n) of points in S converging to s.

One might say this is a reasonable property to expect a topological space to have. However, we will later see examples of spaces that are not Fréchet-Urysohn.

1.8 Hausdorff spaces

Definition 1.8.1. A topological space X is said to be **Hausdorff** or T_2 if any two distinct points can be separated by disjoint neighborhoods: for all $x, y \in X$ with $x \neq y$ there exists an open neighborhood U of x and an open neighborhood V of y such that $U \cap V = \emptyset$.



Many of the spaces that appear naturally in nature are Hausdorff, and many nice theorems in topology require the Hausdorff assumption.

Example 1.8.2. The classical example of a Hausdorff space is $\mathbb{R}^n$ with the standard topology: given any two distinct points x and y, the radius $\varepsilon = \frac{|x-y|}{2}$ determines disjoint open balls

$$x \in B_\varepsilon(x) \quad \text{and} \quad y \in B_\varepsilon(y).$$

Exercise 1.8.3. Show that $\mathbb{R}_\ell$ is a Hausdorff space.

Exercise 1.8.4. Show that $\mathbb{R}$ with the finite complement topology is not a Hausdorff space.

Example 1.8.5. Consider the line with two origins. We showed in Example 1.1.17 that any two open neighborhoods of each of the two origins intersect nontrivially, and thus this space is not Hausdorff.

Lemma 1.8.6. *In a Hausdorff space, every finite subset of X is closed.*

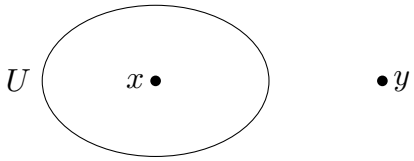
Proof. Let X be a Hausdorff space and fix $x \in X$. Given any $y \in X$ with $y \neq x$, since X is Hausdorff we can find open sets $U \ni x$ and $V_y \ni y$ such that $U \cap V_y = \emptyset$. In particular, $x \notin V_y$. Then

$$\bigcup_{y \neq x} V_y = X \setminus \{x\}$$

is open. We conclude that $\{x\}$ is closed. On the other hand, finite unions of closed sets are closed, and thus any nonempty finite set is closed. □

The proof of Lemma 1.8.6 used less than the Hausdorff condition: we only needed to separate two points by a neighborhood of one that does not contain the other.

Definition 1.8.7. A topological space X is T_1 if for if for all $x, y \in X$ with $x \neq y$ there exists an open neighborhood U of x such that $y \notin U$.



Lemma 1.8.8. *A topological space X is T_1 if and only if every point is closed, meaning that $\{x\}$ is closed for all $x \in X$.*

Proof. ($\Rightarrow$) We essentially already proved this in Lemma 1.8.6: if X is T_1, every one-point set $\{x\}$ is closed since every $y \neq x$ has an open neighborhood V with $x \notin V$, and the finite union of closed sets is closed.

($\Leftarrow$) Given any two points $x \neq y$, since $\{x\}$ is closed then $U = X \setminus \{y\}$ is open, and the open neighborhood U of x does not contain y. □

Remark 1.8.9. Note that every Hausdorff space is T_1.

However, not every T_1 space is Hausdorff:

Example 1.8.10. Let X be an infinite set and consider the finite complement topology on X. Then given any $x \neq y$, the open neighborhood $X \setminus \{y\}$ of x does not contain y, and thus X is T_1. On the other hand, if U is any open neighborhood of x and V is any open neighborhood of Y, their complements are finite, and thus

$$X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$$

is also finite, since it is the union of two finite sets. Thus $U \cap V$ is infinite, and in particular nonempty. Therefore, X is not Hausdorff.

Exercise 1.8.11. Show that the line with two origins is a T_1 space.

The next lemma motivates the name *accumulation point*.

Lemma 1.8.12. *Let X be a T_1 space and $S \subseteq X$. A point $x \in X$ is an accumulation point of S if and only if every open neighborhood of x in X contains infinitely many points of S.*

Proof. ($\Leftarrow$) If an open neighborhood of x intersects S in infinitely many points, then in particular it contains at least one point of S other than x.

($\Rightarrow$) We prove the contrapositive. Assume there is some neighborhood U of x such that $U \cap S$ is finite. Then $C = U \cap (S \setminus \{x\})$ must also finite. Since X is T_1, then $C = U \cap (S \setminus \{x\})$ is closed in X by Lemma 1.8.8. Therefore, its complement

$$X \setminus C = X \setminus (U \cap (S \setminus \{x\}))$$

is open, and hence

$$V := U \cap X \setminus C = U \cap (X \setminus (U \cap (S \setminus \{x\}))) \subseteq U \setminus S \cup \{x\}$$

is open. On the one hand, $x \in U$ and $x \notin C$, so $x \in V$. On the other hand, $V \cap S$ contains at most the one point x, or else it is empty. Thus x is not an accumulation point of S. □

Some of the bad behavior of general topological spaces disappears once the space is Hausdorff. For example, in a Hausdorff space, a sequence can only converge to at most one point (or none).

Theorem 1.8.13. *Let X be a Hausdorff space. Then every sequence of points in X converges to at most one point.*

Proof. Let (x_n) be a sequence of points in X that converges to x. If $y \neq x$, consider open neighborhoods U of x and V of y such that $U \cap V = \emptyset$. Since (x_n) converges to x, there exists N such that $x_n \in U$ for all $n \geqslant N$. Since U and V are disjoint, $x_n \notin V$ for all $n \geqslant N$. We conclude that (x_n) does not converge to y. □

Notation 1.8.14. Let X be a Hausdorff space and (x_n) a sequence of points in X. We write $x_n \longrightarrow x$ to denote that (x_n) converges to x, and say that x is the **limit** of (x_n).

However, the property that every convergent sequence has only one limit does not characterize Hausdorff spaces.

Exercise 1.8.15. Consider following collection of subsets of $\mathbb{R}$:

$$\mathcal{T} = \{U \subseteq \mathbb{R} \mid \mathbb{R} \setminus U \text{ is a countable set } \} \cup \{\emptyset\}.$$

Note that finite sets (including the empty set) are considered to be countable.

a) Show that $\mathcal{T}$ is a topology on $\mathbb{R}$.

b) Let (x_n) be a sequence of real numbers. Show that if (x_n) converges in $(R, \mathcal{T})$, then it must be eventually constant.

c) Show that every convergent sequence in $(\mathbb{R}, \mathcal{T})$ converges to only one limit.

d) Show that $(\mathbb{R}, \mathcal{T})$ is not Hausdorff.

1.9 Continuous functions

Whenever we define a new concept (in this case, a topology) that consists of a set with some additional structure, it is natural to consider what functions between the underlying set behave well with that structure. In the case of topological spaces, the appropriate notion is of a continuous function, a concept which relies on understanding preimages very well.

Given a function $f\colon X \to Y$ and a subset U of Y, recall that the **preimage** of f is

$$f^{-1}(U) = \{x \in X \mid f(x) \in U\}.$$

Before we go on to talk about continuous functions, we record various elementary properties of images and preimages we will use often. More on this topic can be found in Appendix A.

Exercise 1.9.1. Given a function $f\colon X \longrightarrow Y$, show that the following properties hold:

a) For all subsets $A \subseteq X$, $A \subseteq f^{-1}(f(A))$, and equality holds if f is injective.

b) For all subsets $B \subseteq Y$, $f(f^{-1}(B)) \subseteq B$, and equality holds if f is surjective.

c) For all subsets $B_1 \subseteq B_2$ of Y, $f^{-1}(B_1) \subseteq f^{-1}(B_2)$.

d) Given any collection $\{B_\alpha\}_{\alpha\in I}$ of subsets of Y,

$$f^{-1}\left(\bigcup_{\alpha\in I} B_\alpha\right) = \bigcup_{\alpha\in I} f^{-1}\left(B_\alpha\right).$$

e) Given any collection $\{B_\alpha\}_{\alpha\in I}$ of subsets of Y,

$$f^{-1}\left(\bigcap_{\alpha\in I} B_\alpha\right) = \bigcap_{\alpha\in I} f^{-1}\left(B_\alpha\right).$$

f) For all subsets $B_1 \subseteq B_2$ of Y, $f^{-1}(B_2 \setminus B_1) = f^{-1}(B_2) \setminus f^{-1}(B_1)$.

g) For all subsets $A_1 \subseteq A_2$ of X, $f(A_1) \subseteq f(A_2)$.

h) Given any collection of subsets $\{A_\alpha\}_{\alpha\in I}$ of subsets of X,

$$f\left(\bigcup_{\alpha\in I} A_\alpha\right) = \bigcup_{\alpha\in I} f\left(A_\alpha\right).$$

i) Given any collection $\{A_\alpha\}_{\alpha\in I}$ of subsets of X,

$$f\left(\bigcap_{\alpha\in I} A_\alpha\right) \subseteq \bigcap_{\alpha\in I} f\left(A_\alpha\right)$$

with equality if f is injective.

j) For all $A_1 \subseteq A_2 \subseteq X$, $f(A_2) \setminus f(A_1) \subseteq f(A_2 \setminus A_1)$, with equality if f is injective.

Definition 1.9.2. Let X and Y be topological spaces. A function $f\colon X \to Y$ is **continuous** if for every open subset U of Y, its preimage $f^{-1}(U)$ is an open subset of X.

Lemma 1.9.3. *Let X and Y be topological spaces and fix a basis $\mathscr{B}$ for Y. A function $f\colon X \to Y$ is continuous if and only if $f^{-1}(U)$ is open in X for all $U \in \mathscr{B}$.*

Proof. ($\Rightarrow$) is immediate, since basis elements are open in Y.

($\Leftarrow$) Suppose $f^{-1}(U)$ is open in X for all $U \in \mathscr{B}$, and let V be any open subset of Y. By Theorem 1.2.9,

$$V = \bigcup_{\alpha \in I} U_\alpha$$

for some collection $\{U_\alpha\}_{\alpha \in I}$ of basic open sets in $\mathscr{B}$. Then

$$f^{-1}(V) = \bigcup_{\alpha \in I} f^{-1}(U_\alpha)$$

which by assumption is a union of open subsets of X, and hence is open. □

Lemma 1.9.4. *Let X be a topological space and $\mathscr{B}$ be a basis for the topology on X. Let Y be a topological space and $\mathscr{C}$ be a basis for the topology on Y. A function $f\colon X \to Y$ is continuous if and only if for every $x \in X$ and for every $U \in \mathscr{C}$ such that $f(x) \in U$, there exists $V \in \mathscr{B}$ such that $x \in V$ and $f(V) \subseteq U$.*

Proof. ($\Rightarrow$) Suppose f is continuous, and fix $x \in X$ and $U \in \mathscr{C}$ with $f(x) \in U$. By definition of continuous function, $f^{-1}(U)$ is open in X. Moreover, since $f(x) \in U$, then $x \in f^{-1}(U)$. Since $\mathscr{B}$ is a basis for the topology on X, there exists $V \in \mathscr{B}$ with $x \in V \subseteq f^{-1}(U)$, and hence $f(V) \subseteq U$.

($\Leftarrow$) Given any open set W in Y, we need to show that $f^{-1}(W)$ is open in X. Let $x \in f^{-1}(W)$. Then $f(x) \in W$ and so, since $\mathscr{C}$ is a basis for the topology on Y, there is $U \in \mathscr{C}$ such that $f(x) \in U \subseteq W$. By assumption, there exists $V \in \mathscr{B}$ such that $x \in V$ and $f(V) \subseteq U$. Since $U \subseteq W$, we have $V \subseteq f^{-1}(W)$. Thus for all $x \in f^{-1}(W)$ there exists $V \in \mathscr{B}$ such that $x \in V \subseteq f^{-1}(W)$. Since $\mathscr{B}$ is a basis of X, this shows that $f^{-1}(W)$ is open. □

Example 1.9.5. Let X and Y be topological spaces and consider $X \times Y$ under the product topology. The **canonical projection** maps $\pi\colon X \times Y \longrightarrow X$ and $\pi_2\colon X \times Y \longrightarrow Y$ are given by $\pi_1(x, y) = x$ and $\pi_2(x, y) = y$. Given an open set U in X,

$$\pi_1^{-1}(U) = U \times Y$$

is open in $X \times Y$. Thus π_1 is continuous, and similarly one can show that π_2 is continuous.

If Z is another topological space, then given functions $f_1\colon Z \longrightarrow X$ and $f_2\colon Z \longrightarrow Y$, we write $\langle f_1, f_2\rangle\colon Z \longrightarrow X \times Y$ for the function given by

$$\langle f_1, f_2\rangle(z) = (f_1(z), f_2(z)).$$

Note that this is the unique function $Z \longrightarrow X \times Y$ with the property

$$\pi_1 \circ \langle f_1, f_2\rangle = f_1 \qquad \text{and} \qquad \pi_2 \circ \langle f_1, f_2\rangle = f_2.$$

Exercise 1.9.6. Let X, Y, and Z be topological spaces and consider $X \times Y$ under the product topology. Show that if $f_1 \colon Z \longrightarrow X$ and $f_2 \colon Z \longrightarrow Y$ are continuous functions, then the function $\langle f_1, f_2 \rangle \colon Z \longrightarrow X \times Y$ given by

$$\langle f_1, f_2 \rangle(z) = (f_1(z), f_2(z))$$

is also continuous.

We now summarize the content of Example 1.9.5 and Exercise 1.9.6 in the following:

Theorem 1.9.7 (The Universal Property of a Product)**.** *For topological spaces X, Y and Z, let $Y \times Z$ be equipped with the product topology.*

a) Each of the two projection maps $\pi_1 : Y \times Z \to Y$ and $\pi_2 : Y \times Z \to Z$ are continuous.

b) Given continuous maps $g \colon X \to Y$ and $h \colon X \to Z$ there exists a unique continuous map $f \colon X \to Y \times Z$ such that $\pi_1 \circ f = g$ and $\pi_2 \circ f = h$.

Definition 1.9.8. Let X and Y be topological spaces and fix a point $x \in X$. A function $f \colon X \to Y$ is **continuous at** x if for every open neighborhood U of $f(x)$ in Y, there is an open neighborhood V of x in X such that $f(V) \subseteq U$.

Corollary 1.9.9. *Let X and Y be topological spaces. A function $f \colon X \to Y$ is continuous if and only if it is continuous at x for all $x \in X$.*

Proof. This follows from Lemma 1.9.4 by taking $\mathscr{B}$ to be all open sets in X and $\mathscr{C}$ to be the all open sets in Y. □

Corollary 1.9.10. *Consider $\mathbb{R}$ under the standard topology. A function $f \colon \mathbb{R} \to \mathbb{R}$ is continuous if and only if for every point $r \in \mathbb{R}$ and every $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$|x - r| < \delta \implies |f(x) - f(r)| < \varepsilon.$$

Remark 1.9.11. One might think that Corollary 1.9.10 is an immediate consequence of Lemma 1.9.4 by using the basis

$$\mathscr{B} = \{(a, b) \mid a < b\}$$

on both the domain and codomain. But this is not quite right, since in the statement of Corollary 1.9.10 we restrict to using intervals *centered at* the points r and $f(r)$.

Proof. Since

$$|x - r| < \delta \iff x \in (r - \delta, r + \delta),$$

we want to show that f is continuous if and only if for every $r \in \mathbb{R}$ and every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$f((r - \delta, r + \delta)) \subseteq (f(r) - \varepsilon, f(r) + \varepsilon).$$

Thanks to Lemma 1.9.4, it suffices to show that this condition is equivalent to the following: taking the basis for $\mathbb{R}$ given by

$$\mathscr{B} = \{(a, b) \mid a < b\}$$

on both domain and codomain, for all intervals (a, b) with $f(r) \in (a, b)$, there exists $\varepsilon > 0$ such that $(f(r) - \varepsilon, f(r) + \varepsilon) \subseteq (a, b)$.

Fix $r \in \mathbb{R}$, and suppose that for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$|x - r| < \delta \implies |f(x) - f(r)| < \varepsilon.$$

Pick any open interval (a, b) with $f(r) \in (a, b)$. Note that $a < f(r) < b$, so

$$\varepsilon := \min\{f(r) - a, b - f(r)\} > 0 \qquad \text{and} \qquad (f(r) - \varepsilon, f(r) + \varepsilon) \subseteq (a, b).$$

By assumption, there exists $\delta > 0$ such that

$$f((r - \delta, r + \delta)) \subseteq (f(r) - \varepsilon, f(r) + \varepsilon) \subseteq (a, b).$$

By Lemma 1.9.4, f is continuous.

Conversely, assume for all open intervals (a, b) with $f(r) \in (a, b)$, there exists $\varepsilon > 0$ such that

$$(f(r) - \varepsilon, f(r) + \varepsilon) \subseteq (a, b).$$

Given $r \in \mathbb{R}$ and $\varepsilon > 0$, $(f(r) - \varepsilon, f(r) + \varepsilon)$ is an open interval containing r. Thus by assumption, there is some open interval (a, b) with $r \in (a, b)$ and

$$f((a, b)) \subseteq (f(r) - \varepsilon, f(r) + \varepsilon).$$

Taking $\delta := \min\{r - a, b - r\}$, we now have

$$r \in (r - \delta, r + \delta) \subseteq (a, b) \qquad \text{and hence} \qquad f((r - \delta, r + \delta)) \subseteq (f(r) - \varepsilon, f(r) + \varepsilon). \quad \square$$

Theorem 1.9.12. *Let X and Y be topological spaces. The following conditions on a function $f \colon X \to Y$ are equivalent:*

a) The function f is continuous.

b) The function f is continuous at x for all $x \in X$.

c) For all C closed in Y, $f^{-1}(C)$ is closed in X .

d) Every subset S of X satisfies $f(\overline{S}) \subseteq \overline{f(S)}$.

Proof. The equivalence of a and b was proved in Corollary 1.9.9.

a $\iff$ c: For all $S \subseteq Y$, by Exercise 1.9.1 we have

$$X \setminus f^{-1}(S) = f^{-1}(X \setminus S).$$

a $\implies$ d: Assume f is continuous and let $S \subseteq X$. Given $x \in \overline{S}$, by Theorem 1.7.7 every open neighborhood of x in X intersects S nontrivially. Let V be an open neighborhood of $f(x)$.

Since f is continuous, $f^{-1}(V)$ is an open neighborhood of x in X. Since $x \in \overline{S}$, $f^{-1}(V) \cap S$ is nonempty. Therefore, $V \cap f(S)$ is nonempty. By Theorem 1.7.7, $f(x) \in \overline{S}$. This shows that $f(\overline{S}) \subseteq \overline{f(S)}$.

d $\implies$ c: Let C be closed in Y. Then

$$\begin{aligned} f(\overline{f^{-1}(C)}) &\subseteq \overline{f(f^{-1}(C)} && \text{by d} \\ &\subseteq \overline{C} && \text{applying Exercise 1.7.6 to } f(f^{-1}(C) \subseteq C \text{ from Exercise 1.9.1} \\ &= C && \text{since } C \text{ is closed.} \end{aligned}$$

Therefore, $\overline{f^{-1}(C)} \subseteq f^{-1}(C)$. Since $f^{-1}(C)$ contains its own closure, it must be closed. □

Example 1.9.13. For any continuous function $f\colon \mathbb{R} \longrightarrow \mathbb{R}$, we have $f(1) \in \overline{f([0,1))}$ since $1 \in \overline{[0,1)}$.

Example 1.9.14. Consider $\mathbb{R}$ under the standard topology. Consider any continuous function $f\colon \mathbb{R} \longrightarrow \mathbb{R}$, where both copies of $\mathbb{R}$ are given the standard topology, and

$$K = \left\{ \frac{1}{n} \mid n \in \mathbb{Z}, n \geqslant 1 \right\}.$$

By Exercise 1.7.19, $\overline{K} = K \cup \{0\}$. Therefore,

$$f(0) \subseteq \overline{\left\{ f\left(\frac{1}{n}\right) \mid n \in \mathbb{Z}, n \geqslant 1 \right\}}.$$

Remark 1.9.15. In general, the image of an open set by a continuous function need not be an open set, and the image of a closed set need not be a closed set. For example, we will see below in Lemma 1.9.19 that all constant functions are continuous, so the constant function $f\colon \mathbb{R} \longrightarrow \mathbb{R}$ equal to zero is continuous in the standard topology. However, $f(\mathbb{R}) = \{0\}$ is not open.

Exercise 1.9.16. Give an example of a continuous function $f\colon X \longrightarrow Y$ and a closed set C in X such that $f(C)$ is not closed.

Definition 1.9.17. Let $f\colon X \to Y$ be a function between topological spaces. We say f is an **open map** if

$$U \text{ open in } X \implies f(U) \text{ open in } Y.$$

We saw f is a **closed map** if

$$C \text{ closed in } X \implies f(C) \text{ closed in } Y.$$

Exercise 1.9.18. Let X and Y be topological spaces and endow $X \times Y$ with the product topology.

a) Show that the projection maps $\pi_1\colon X \times Y \longrightarrow X$ and $\pi_2\colon X \times Y \longrightarrow Y$ are open.

b) Projection maps need not be closed: endowing $\mathbb{R}$ with the standard topology, show that the canonical projection $\pi_1\colon \mathbb{R}^2 \longrightarrow \mathbb{R}$ is not closed.

Lemma 1.9.19. *Continuous functions enjoy the following properties:*

a) The composition of any two continuous maps is again continuous: given topological spaces X, Y, and Z and continuous functions $f\colon X \to Y$ and $g\colon Y \to Z$, the composition $g \circ f\colon X \to Z$ is continuous.

b) The inclusion of a subspace into a space is a continuous function.

c) If $f\colon X \to Y$ is continuous and A is subspace of X, then the restriction $f|_A\colon A \to Y$ to A is continuous.

d) If $f\colon X \to Y$ is continuous, and $f(X) \subseteq Z$ for some subspace Z of Y, then $f|^Z\colon X \to Z$ is continuous.

e) Every constant map is continuous.

Proof.

a) Given any subset U of Z, we have

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)).$$

If U is open in Z, then $g^{-1}(U)$ is open in Y since g is continuous. Since f is continuous, then $f^{-1}(g^{-1}(U))$ is open.

b) Given a subspace Z of Y, let $\iota\colon Z \longrightarrow Y$ be the inclusion map. Then for any subset U of Y, $\iota^{-1}(U) = U \cap Z$. In particular, if U is open in Y then $\iota^{-1}(U)$ is open in Z by the definition of the subspace topology.

c) We can view the restriction of f to A as a composition of the inclusion of A into X with the original function f. Since the composition of continuous functions is continuous, then the restriction is continuous.

d) Let $V \subseteq Y$. Since $f(X) \subseteq Z$,

$$f(x) \in V \cap Z \iff f(x) \in V.$$

Fix an open set U in Z. Then $U = V \cap Z$ for some V open in Y, and

$$\left(f|^Z\right)^{-1}(U) = \{x \in X \mid f(x) \in V \cap Z\} = \{x \in X \mid f(x) \in V\} = f^{-1}(V)$$

is open in X since f is continuous.

e) Consider the constant map $f\colon X \to Y$ with $f(x) = c$ for all $x \in X$. For all open sets U in Y,

$$f^{-1}(U) = \begin{cases} \emptyset & \text{if } c \notin U \\ X & \text{if } c \in U. \end{cases}$$

Either way, $f^{-1}(U)$ is open. □

Theorem 1.9.20 (Pasting lemma for open subsets)**.** *Fix an open cover $\{U_\alpha\}_{\alpha\in I}$ of a topological space X, meaning a collection of open sets U_α with*

$$X = \bigcup_{\alpha\in I} U_\alpha.$$

A function $f\colon X \to Y$ is continuous if and only if $f|_{U_\alpha} : U_\alpha \to Y$ is continuous for all $\alpha \in I$.

Proof. The implication ($\Rightarrow$) is a special case of Lemma 1.9.19 c: restricting the domain of a continuous function gives a continuous function.

($\Leftarrow$) Fix an open cover $\{U_\alpha\}$ for X and assume $f|_{U_\alpha} : U_\alpha \to Y$ is continuous for all $\alpha \in I$. Let V be open in Y. Then

$$f^{-1}(V) = \bigcup_{\alpha\in I}(f^{-1}(V)\cap U_\alpha) = \bigcup_{\alpha\in I}\left(f|_{U_\alpha}^{-1}(V)\right)$$

is open in X. □

Remark 1.9.21. As a consequence of Theorem 1.9.20, we can glue continuous maps along open subsets: given an open cover $\{U_\alpha\}_{\alpha\in I}$ of X, suppose we are given a continuous map $g_\alpha : U_\alpha \to Y$ for each $\alpha \in I$, and that for all α, β we have that g_α and g_β agree on $U_\alpha \cap U_\beta$. We then set $f\colon X \to Y$ by $f(x) = g_\alpha(x)$ whenever $x \in U_\alpha$. Note that this makes sense thanks to the assumptions: since $\{U_\alpha\}_{\alpha\in I}$ is an open cover for X, every $x \in X$ belongs to at least one U_α, and f is well-defined since $x \in U_\alpha \cap U_\beta$ implies

$$g_\alpha(x) = g_\beta(x).$$

Then by Theorem 1.9.20, f is continuous, since $f|_{U_\alpha} = g_\alpha$ is continuous for all $\alpha \in I$.

We can also glue for *finite* collections of closed subsets.

Lemma 1.9.22 (Pasting lemma for closed subsets)**.** *Let X and Y be topological spaces. Let A and B be closed subsets of X such that $A\cup B = X$. Given continuous maps $g\colon A \to Y$ and $h\colon B \to Y$ such that $g(x) = h(x)$ for all $x \in A \cap B$, there is a unique continuous function $f\colon X \to Y$ such that $f|_A = g$ and $f|_B = h$.*

Proof. Define $f\colon X \to Y$ by $f(x) = g(x)$ if $x \in A$ and $f(x) = h(x)$ if $x \in B$. This is well-defined since $g(x) = h(x)$ for all $x \in A \cap B$. Note that f is the only function such that $f|_A = g$ and $f|_B = h$. We need to prove f is continuous.

Let C be a closed subset of Y. Then

$$f^{-1}(C) = g^{-1}(C) \cup h^{-1}(C).$$

Since g is continuous, $g^{-1}(C)$ is closed in A by Theorem 1.9.12. Since A is closed in X, then $g^{-1}(C)$ is closed in X by Remark 1.6.7. Similarly, $h^{-1}(C)$ is closed in X. Since *finite* unions of closed subsets are closed, it follows that $f^{-1}(C)$ is closed in X. By Theorem 1.9.12, this proves f is continuous. □

Remark 1.9.23. We can generalize Lemma 1.9.22 to any finite collection of open sets that cover X: if $A_1, \dots, A_n$ are closed sets such that $A_1 \cup \cdots \cup A_n = X$, then any continuous maps $g_i \colon A_i \longrightarrow Y$ with

$$g_i|_{A_i \cap A_j} = g_j|_{A_i \cap A_j} \qquad \text{for all } i, j$$

can be glued to make a continuous map $g \colon X \longrightarrow Y$ defined by

$$g(x) = g_i(x) \qquad \text{if } x \in A_i.$$

To prove this, one can proceed inductively, extending from A_1 to $A_1 \cup A_2$ and so on, up until $A_1 \cup \cdots \cup A_n = X$.

Exercise 1.9.24. Show that Lemma 1.9.22 does not generalize to an arbitrary collection of closed sets of X with

$$\bigcup_i A_i = X.$$

Example 1.9.25. Consider the functions $g \colon A := (-\infty, 0] \to \mathbb{R}$ and $h \colon B := [0, \infty) \to \mathbb{R}$ given by

$$g(x) = 0 \text{ for all } x \in A \qquad \text{and} \qquad h(x) = x^2 \text{ for all } x \in B.$$

The functions g and h are continuous, which we can justify by proving they are restrictions of continuous functions $\mathbb{R} \to \mathbb{R}$ to subspaces of $\mathbb{R}$. Moreover, g and h agree on $A \cap B = \{0\}$, and so by Lemma 1.9.22 they glue to give the continuous function $f \colon \mathbb{R} \to \mathbb{R}$ given by

$$f(x) = \begin{cases} 0 & \text{if } x \leqslant 0 \\ x^2 & \text{if } x \geqslant 0. \end{cases}$$

Definition 1.9.26. Let X be a topological space and consider $A \subseteq X$ under the subspace topology. A **retraction** of X into A is a continuous function $r \colon X \longrightarrow A$ such that $r(a) = a$ for all $a \in A$.

Exercise 1.9.27. Let X be a Hausdorff space.

a) Show that if $f \colon X \to X$ is continuous, then

$$C := \{x \in X \mid f(x) = x\}$$

is closed.

b) Let $A \subseteq X$ have the subspace topology. Show that if there exists a retraction of X into A, then A is a closed subset of X.

c) Does (b) hold if we do not assume that X is Hausdorff?

Continuous functions always take convergent sequences to convergent sequences.

Exercise 1.9.28. Let X and Y be topological spaces and let $f \colon X \longrightarrow Y$ be a continuous functions. Show that if f is continuous, then f sends convergent sequences to convergent sequences, meaning that for every sequence (x_n) in X converging to x, the sequence $(f(x_n))$ converges to $f(x)$.

The converse holds if X is Fréchet-Urysohn.

Exercise 1.9.29. Show that if X is Fréchet-Urysohn and $f \colon X \longrightarrow Y$ sends convergent sequences to convergent sequences, then f is continuous.

1.10 Homeomorphisms

Definition 1.10.1. Let X and Y be topological spaces. A **homeomorphism** from X to Y is a bijection $f\colon X \longrightarrow Y$ such that both f and f^{-1} are continuous. If such a map exists, we say X and Y are **homeomorphic**.

Remark 1.10.2. If f is a homeomorphism, then so is its inverse function f^{-1}.

However, here is a very important (and perhaps surprising) warning: a continuous bijection need not be a homeomorphism.

Example 1.10.3. Let S^1 denote the unit circle in $\mathbb{R}^2$, endowed with the subspace topology induced by the standard topology on $\mathbb{R}^2$. Let $f\colon [0,1) \longrightarrow S^1$ be the map given by

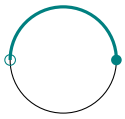
$$f(t) = (\cos(2\pi t), \sin(2\pi t)).$$

One can check that f is bijective and continuous, but the inverse map is *not* continuous, and so f is *not* a homeomorphism.

To see the inverse is not continuous, consider

$$(f^{-1})^{-1}([0,1/2)) = f([0,1/2)).$$

There is no open subset of $\mathbb{R}^2$ whose intersection with S^1 is this set, while $[0,1/2)$ is an open set in $[0,1)$.



Two homeomorphic topological spaces are indistinguishable *from a topological point of view*. In other words, homeomorphism is the appropriate notion of isomorphism for topological spaces.

Example 1.10.4. The function $f \longrightarrow (-1,1) \longrightarrow \mathbb{R}$ given by $f(x) = \frac{x}{1-x^2}$ is a bijection. If we endow $\mathbb{R}$ with the standard topology, and $(-1,1)$ under the subspace topology, the function f is continuous. One can justify this by using results from calculus and Corollary 1.9.10. Its inverse $f^{-1}\colon \mathbb{R} \to (-1,1)$ is given by

$$f^{-1}(y) = \frac{2y}{1+\sqrt{1+4y^2}},$$

which is also continuous, and again we can justify this using results from analysis. Therefore, f is a homeomorphism, and $(-1,1)$ and $\mathbb{R}$ are homeomorphic topological spaces.

Remark 1.10.5. If $f\colon X \to Y$ is a bijection, then for any subset U of Y, the preimage of U and the image of U under f^{-1} are the same. We may write this in a very unhelpful way as

$$f^{-1}(U) = f^{-1}(U).$$

This partially explains the notation for preimages we have been using.

Moreover, $(f^{-1})^{-1}(U) = f(U)$ in any way you would like to interpret this, using preimages or the image under the inverse function. Finally, $f^{-1}(f(U)) = U$.

Lemma 1.10.6. *Let $f\colon X \longrightarrow Y$ be a bijection between two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$. The following are equivalent:*

a) The function f is a homeomorphism.

b) A subset U of X is open in X if and only if $f(U)$ is open in Y.

c) We have a bijection

$$\begin{aligned} \mathcal{T}_X &\longleftrightarrow \mathcal{T}_Y \\ U &\longmapsto f(U) \\ f^{-1}(V) &\longmapsfrom V. \end{aligned}$$

Proof. a $\Rightarrow$ b: Let f be a homeomorphism and U be a subset of X. If $f(U)$ is open in Y then $U = f^{-1}(f(U))$ is open in X since f is continuous. If U is open in X, then $f(U) = (f^{-1})^{-1}(U)$ is open in Y since $f^{-1}(U)$ is continuous.

b $\Rightarrow$ a: For all V open in Y,

$$f(f^{-1}(V)) = V$$

is open in Y, and thus $f^{-1}(V)$ is open in X. Therefore, f is continuos. For all open subsets U of X, we have

$$(f^{-1})^{-1}(U) = f(U)$$

which is open in Y by assumption; this proves that f^{-1} is continuous. We conclude that f is a homeomorphism.

b $\Rightarrow$ c: The assumption guarantees that the assignment $U \mapsto f(U)$ determines a function $\mathcal{T}_X \mapsto \mathcal{T}_Y$. Since f is bijective, for all open sets $V \in \mathcal{T}_Y$, the set $U = f^{-1}(V)$ satisfies

$$f(U) = f(f^{-1}(V)) = V.$$

Since V is open, by b we conclude that $U = f^{-1}(V)$ is open. Thus $V \mapsto f^{-1}(V)$ defines a function $\mathcal{T}_Y \longrightarrow \mathcal{T}_X$. Finally, since f is bijective then the assignments $U \mapsto f(U)$ and $V \mapsto f^{-1}(V)$ are inverse to each other.

c $\Rightarrow$ b: By c, if $U \in \mathcal{T}_X$ then $f(U) \in \mathcal{T}_Y$. Conversely, if $V = f(U) \in \mathcal{T}_Y$, then

$$f^{-1}(V) = f^{-1}(f(U)) \in \mathcal{T}_X.$$

These assignments are inverse to each other, so $U = f^{-1}(f(U)) \in \mathcal{T}_X$. □

Homeomorphic spaces should be thought of as essentially the same, at least from the point of view of topology. Any meaningful topological property should be invariant under homeomorphisms. We call such properties topological invariants. More precisely:

Definition 1.10.7. A property P of topological spaces is a **topological invariant** if for all homeomorphisms $f\colon X \longrightarrow Y$, the space X has property P if and only if Y has property P.

A *good* topological property is one that is topologically invariant. Here are some examples:

Example 1.10.8.

a) The property of being T_2 is a topological invariant.

b) The property of being T_1 is a topological invariant.

c) Having a countable basis is a topological invariant.

d) Homeomorphisms need not preserve any sort of distance notion. For instance, we saw in Example 1.10.4 that $\mathbb{R}$ and $(-1, 1)$ are homeomorphic, but each has a standard notion of distance, and while the distance in $(-1, 1)$ is bounded, the distance in $\mathbb{R}$ is not.

Exercise 1.10.9. Let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two topologies on X. Prove that the identify function $\mathrm{id}\colon (X, \mathcal{T}_1) \longrightarrow (X, \mathcal{T}_2)$ is continuous if and only if $\mathcal{T}_2$ is finer than $\mathcal{T}_1$.

Lemma 1.10.10. *Here are some properties of homeomorphisms:*

a) The identity function $\mathrm{id}\colon X \longrightarrow X$*, where both domain and codomain are given the same topology, is a homeomorphism.*

b) The inverse of a homeomorphism is a homeomorphism.

c) The composition of two homeomorphisms is again a homeomorphism.

Proof.

a) By Lemma 1.10.6, it suffices to see that $U \mapsto U$ determines a bijection on the open sets of X.

b) If f is a homeomorphism, then its inverse f^{-1} is a continuous function by definition of homeomorphism, and its inverse is f itself, which is continuous by assumption.

c) Let $f\colon X \longrightarrow Y$ and $g\colon Y \longrightarrow Z$ be homeomorphisms. The composition of bijections is a bijection, so gf is a bijective. By Lemma 1.9.19, the composition of continuous functions is continuous, so gf is continuous. Finally, $(gf)^{-1} = g^{-1}f^{-1}$ is a composition of continuous bijections, and thus a continuous bijection. □

Corollary 1.10.11. *The relation of being homeomorphic is an equivalence relation.*

Proof. This follows from Lemma 1.10.10: given topological spaces X, Y, and Z,

- Since the identity on X is a homeomorphism, X is homeomorphic to itself.
- If X is homeomorphic to Y and $f\colon X \longrightarrow Y$ is a homeomorphism, then Y is homeomorphic to X via f^{-1}.
- If X is homeomorphic to Y via the homeomorphism $f\colon X \longrightarrow Y$ and Y is homeomorphic to Z via the homeomorphism $g\colon Y \longrightarrow Z$, then X is homeomorphic to Z via the homeomorphism gf.

□

Definition 1.10.12. An **embedding** is a homeomorphism onto its image, that is, an injective map $f\colon X \longrightarrow Y$ such that the restriction $X \longrightarrow f(X)$ is a homeomorphism.

1.11 The product topology: the general case

Given an arbitrary collection of topological spaces, there are a few natural ways to define a topology on the cartesian product of the underlying sets that is induced by the topologies on each individual space.

Exercise 1.11.1. Let $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$ be an arbitrary collection of topological spaces indexed by some indexing set I. Show that the set

$$\mathscr{B} = \left\{ \prod_{\alpha \in I} U_\alpha \mid U_\alpha \in \mathcal{T}_\alpha \right\}$$

is a basis for a topology on

$$\prod_{\alpha \in I} X_\alpha.$$

Definition 1.11.2. The **box topology** on the cartesian product of an arbitrary collection $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha \in I}$ of topological spaces indexed by a set I is the topology with basis

$$\left\{ \prod_{\alpha \in I} U_\alpha \mid U_\alpha \in \mathcal{T}_\alpha \right\}.$$

At first, one might think this is the most natural generalization is the definition of the product topology on two spaces. It turns out, there is a *better* topology on the product. To motivate it, we look back at the case of two spaces.

Discussion 1.11.3. We saw in Example 1.9.5 that given two topological spaces $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$, the product topology on $X \times Y$ makes the projection maps

$$\begin{aligned} \pi_1 \colon X \times Y &\longrightarrow X & \qquad \pi_2 \colon X \times Y &\longrightarrow Y \\ (x, y) &\longmapsto x & (x, y) &\longmapsto y \end{aligned}$$

continuous. In fact, more is true: we claim that this is the unique coarsest topology on $X \times Y$ that makes π_1 and π_2 continuous. To see this, consider any topology $\mathcal{T}_{X \times Y}$ on $X \times Y$. Note that if π_1 and π_2 are both continuous for this topology on $X \times Y$, then for all open sets $U \in \mathcal{T}_X$ and $V \in \mathcal{T}_Y$

$$\pi^{-1}(U) = U \times Y \qquad \text{and} \qquad \pi^{-1}(V) = X \times V$$

must both be open in $X \times Y$. Moreover, since $\mathcal{T}_{X \times Y}$ is a topology, then

$$U \times V = (U \times Y) \cap (X \times V) = U \times V \in \mathcal{T}_{X \times Y}.$$

This shows that $\mathcal{T}_{X \times Y}$ must be finer than the product topology on $X \times Y$. On the other hand, π_1 and π_2 are both continuous for the product topology on $X \times Y$. We conclude that the product topology on $X \times Y$ is the coarsest topology for which π_1 and π_2 are both continuous.

Now consider an arbitrary family of topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha\in I}$ indexed by a set I. Let $\mathcal{T}$ be any topology on the cartesian product for which all projection maps

$$\begin{aligned} \pi_\beta \colon \prod_{\alpha\in I} X_\alpha &\longrightarrow X_\beta \\ (x_\alpha)_{\alpha\in I} &\longmapsto x_\beta \end{aligned}$$

are continuous. By the same argument as above, $\mathcal{T}$ must be such that for all open sets $U_\beta \in \mathcal{T}_\beta$

$$\pi_\beta^{-1}(U_\beta) \in \mathcal{T}.$$

Now note that

$$\pi_\beta^{-1}(U_\beta) = \prod_{\alpha\in I} A_\alpha \qquad \text{where } A_\alpha = \begin{cases} U_\beta & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta. \end{cases}$$

To simplify notation, for a fixed $\beta \in I$ and $U_\beta \in \mathcal{T}_\beta$ write

$$P(\beta, U_\beta) \coloneqq \prod_{\alpha\in I} P_\alpha \qquad \text{where } P_\alpha = \begin{cases} U_\beta & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta. \end{cases}$$

In fact, the set

$$\left\{P(\beta, U_\beta) \mid \beta \in I, U_\beta \in \mathcal{T}_\beta\right\}$$

forms a subbasis for the cartesian product: given any point

$$(x_\alpha)_{\alpha\in I} \in \prod_{\alpha\in I} X_\alpha$$

in the cartesian product, we can fix $\beta \in I$ and find an open neighborhood $U_\beta \in \mathcal{T}_\beta$ of x_β in X_β, and

$$(x_\alpha)_{\alpha\in I} \in P(\beta, U_\beta) = \prod_{\alpha\in I} A_\alpha \qquad \text{where } A_\alpha = \begin{cases} U_\beta & \text{if } \alpha = \beta \\ X_\alpha & \text{if } \alpha \neq \beta. \end{cases}$$

Recall (see Exercise 1.2.27) that any subbasis determines a basis, by taking all the finite intersection of subbasis elements. Given $\beta_1, \dots, \beta_n \in I$ and open sets $U_{\beta_i} \in \mathcal{T}_{\beta_i}$, we have

$$P(\beta_1, U_{\beta_1}) \cap \cdots \cap P(\beta_n, U_{\beta_n}) = \prod_{\alpha\in I} B_\alpha \quad \text{with } B_\alpha = \begin{cases} U_{\beta_1} & \text{if } \alpha = \beta_1 \\ \vdots & \\ U_{\beta_n} & \text{if } \alpha = \beta_n \\ X_\alpha & \text{if } \alpha \notin \{\beta_1, \dots, \beta_n\}. \end{cases}$$

Thus the set

$$\left\{ \prod_{\alpha\in I} U_\alpha \mid U_\alpha \in T_\alpha \text{ with } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in I \right\}$$

is a basis for a topology on the cartesian product, which we call the product topology, and this is the coarsest topology that makes all the projection maps continuous.

In fact, more generally, there is always a unique coarsest topology that makes a particular function continuous:

Lemma 1.11.4. *Let $f\colon S \longrightarrow Z$ be a function with Z a topological space. There is a unique coarsest topology on S that makes f continuous. More generally, given a collection of topological spaces $\{Z_\beta\}_{\beta\in L}$ and functions $f_\beta\colon S \longrightarrow Z_\beta$, there is unique coarsest topology on S so that $f_\beta\colon S \longrightarrow Z_\beta$ is a continuous function for all β.*

Proof. Consider the collection $\{\mathcal{T}_\gamma\}_{\gamma\in I}$ of all the topologies that make f continuous. Note that this is a nonempty collection of topologies: f is always continuous under the discrete topology on S, since the preimage of any open set is a subset of S, and thus open. By Exercise 1.1.28,

$$\mathcal{T} := \bigcap_{\gamma\in I} \mathcal{T}_\gamma$$

is a topology on S. Moreover, for any open subset U of Z, we have $f^{-1}(U) \in \mathcal{T}_\gamma$ for all $\gamma \in I$, and hence $f^{-1}(U) \in \mathcal{T}$. Thus $\mathcal{T}$ makes f continuous, and moreover if $\mathcal{T}_\gamma$ is any topology such that f is continuous, then by definition $\mathcal{T} \subseteq \mathcal{T}_\gamma$.

Now let $\mathcal{T}_\beta$ be the coarsest topology on S making f_β continuous, and consider

$$G := \bigcup_{\beta\in L} \mathcal{T}_\beta.$$

While G is not necessarily a topology, it is a subbasis for a topology: since each $\mathcal{T}_\beta$ is a topology, $S \in \mathcal{T}_\beta$ for all $\beta \in L$, and thus $S \in G$. Now note that any topology on S for which every f_β is continuous must contain $\mathcal{T}_\beta$ for all $\beta \in L$, and thus it must contain G and the topology it generates. On the other hand, the topology generated by G is a topology containing every T_β, and thus it is both a topology for which all the f_β are continuous and in fact the coarsest of them all. □

We are thus ready to define the product topology on an arbitrary product.

Definition 1.11.5. Given an arbitrary family of topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha\in I}$ indexed by a set I, the **product topology** or **Tychonoff topology** on the cartesian product

$$\prod_{\alpha\in I} X_\alpha$$

is the topology with basis

$$\left\{ \prod_{\alpha\in I} U_\alpha \mid U_\alpha \in T_\alpha \text{ with } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in I \right\}.$$

Remark 1.11.6. Note that when I is finite, the Tychonoff topology on

$$\prod_{\alpha\in I} X_\alpha$$

is the usual product topology we discussed in Section 1.4, and it coincides with the box topology.

When taking the product of infinitely many sets, the Tychonoff topology is strictly coarser.

Example 1.11.7. Consider the product of countably many copies of $\mathbb{R}$:

$$\prod_{n\geqslant 1} \mathbb{R}.$$

Give each copy of $\mathbb{R}$ the standard topology, and equip $\mathbb{R}^{\mathbb{N}}$ the product topology. The following collection of open subsets forms a basis for $\mathbb{R}^{\mathbb{N}}$ in the product topology:

$$(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n) \times \prod_{j\geqslant n+1} \mathbb{R}.$$

Strictly speaking, this is a smaller basis than the one we described in Discussion 1.11.3 and in the definition.

The set

$$\prod_{n\geqslant 0} (0, 1)$$

is open in the box topology but not in the Tychonoff topology. Indeed, if it were open in the box topology, then it would contain a nonempty open set of the form

$$U = \prod_{n\geqslant 0} U_\alpha$$

with $U_n = \mathbb{R}$ for all but finitely many values of n. In particular, there exist β such that $U_\beta = \mathbb{R}$, and we could then choose any $x_\alpha \in U_\alpha$ for $\alpha \neq \beta$ and $x_\beta = 2 \in U_\beta = \mathbb{R}$, so that

$$x = (x_\alpha)_{\alpha\geqslant 0} \in \prod_{\alpha\in I} U_\alpha = U \qquad \text{but} \qquad x \notin \prod_{n\geqslant 0} (0, 1).$$

Notation 1.11.8. We will often write the product of countably many copies of $\mathbb{R}$ as

$$\mathbb{R}^\omega = \prod_{n=1}^{\infty} \mathbb{R}.$$

Remark 1.11.9. We can interpret the set $\mathbb{R}^\omega$ as the set of all sequences of real numbers: an element in $\mathbb{R}^\omega$ is a sequence

$$(x_n)_{n\geqslant 1}.$$

Exercise 1.11.10. Show that $\mathbb{R}^\omega$ has a countable basis under the product topology.

While we have given some motivation for why one might prefer the Tychonoff topology to the box topology, as we learn more about desirable topological properties we shall become increasingly convinced that the Tychonoff topology is strictly better than the box topology. In particular, we will see that many nice properties on the individual spaces pass on to the product topology but not always to the box topology.

The following is the general version of Theorem 1.9.7. This is one way in which the Tychonoff topology is better.

Theorem 1.11.11 (Universal Property of the Product)**.** *Consider a topological space Y, a family of topological spaces $\{(X_\alpha, \mathcal{T}_\alpha)\}_{\alpha\in I}$, and a family of functions $f_\alpha \colon Y \longrightarrow X_\alpha$. Consider the function*

$$\begin{aligned} f \colon Y &\longrightarrow \prod_{\alpha\in I} X_\alpha \\ y &\longmapsto (f_\alpha(y))_{\alpha\in I}. \end{aligned}$$

Under the product topology on the Cartesian product, f is continuous if and only if f_α is continuous for all $\alpha \in I$.

Proof. ($\Rightarrow$) Let

$$\pi_\beta \colon \prod_{\alpha\in I} \longrightarrow X_\beta$$

denote the projection onto X_β. By construction of the product topology, each π_α is continuous. If f is continuous, then $f_\alpha = \pi_\alpha \circ f$ is continuous for all $\alpha \in I$ by Lemma 1.9.19, since it is a composition of continuous functions.

($\Leftarrow$) Suppose that f_α is continuous for all α. By Lemma 1.9.4, it suffices to check $f^{-1}(U)$ is open for every open set in a basis for the product topology. Thus it suffices to consider open sets of the form

$$U = \prod_{\alpha\in I} U_\alpha \qquad \text{with } U_\alpha = X_\alpha \text{ for all but finitely many } \alpha \in I.$$

Suppose that

$$\{\alpha \in I \mid U_\alpha \neq X_\alpha\} = \{\alpha_1, \dots, \alpha_n\} =: J.$$

Now note that

$$s \in f^{-1}(U_\alpha) \iff f(s) = (f_\alpha(s))_{\alpha\in I} \in U \iff f_\alpha(s) \in U_\alpha \text{ for all } \alpha \in I.$$

When $\alpha \notin J$, $f_\alpha(s) \in U_\alpha = X_\alpha$ is automatically satisfied. Thus

$$s \in f^{-1}(U_\alpha) \iff f_{\alpha_i}(s) \in U_{\alpha_i} \text{ for } i = 1, \dots, n.$$

Thus

$$f^{-1}(U_\alpha) = \bigcap_{i=1}^{n} f_{\alpha_i}(U_{\alpha_i})$$

is open since it is an intersection of finitely many open sets, as each f_α is continuous and each U_α is open in X_α. □

Remark 1.11.12. In Theorem 1.11.11, the proof of ($\Rightarrow$) used only the fact that the projection maps are continuous in the product topology, and thus this direction holds for any topology with that property. In particular, it holds for the box topology.

However, the other implication is false in the box topology:

Example 1.11.13. Consider the diagonal function

$$\begin{aligned} \Delta\colon \mathbb{R} &\longrightarrow \mathbb{R}^\omega \\ t &\longmapsto (t) = (t, t, t, ...). \end{aligned}$$

We can think of Δ as $\Delta = (\mathrm{id})_{n\geqslant 1}$, where we take the identity function $\mathrm{id}\colon \mathbb{R} \longrightarrow \mathbb{R}$ on each copy of $\mathbb{R}$. If we endow each copy of $\mathbb{R}$ within sight with the standard topology, then Δ is continuous under the product topology on $\mathbb{R}^\omega$ by Theorem 1.11.11. However, we claim that Δ is not continuous if instead we endow $\mathbb{R}^\omega$ with the box topology. Indeed, the set

$$U = \prod_{i=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right)$$

is open in the box topology, but

$$\Delta^{-1}(U) = \bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$$

is not open in $\mathbb{R}$ under the standard topology.

Exercise 1.11.14. Let $\{X_\alpha\}_{\alpha\in I}$ be any collection of topological spaces and endow

$$\prod_\alpha X_\alpha$$

with the product topology. Let A_α be a subset of X_α for each α.

a) Show that if every X_α is Hausdorff, then the product $\prod_{\alpha\in I} X_\alpha$ is Hausdorff.

b) Show that if every X_α is T_1, then the product $\prod_{\alpha\in I} X_\alpha$ is T_1.

c) Show that

$$\prod_{\alpha\in I} \overline{A_\alpha} = \overline{\prod_{\alpha\in I} A_\alpha}.$$

d) Consider

$$A = \prod_\alpha A_\alpha.$$

Show that the product topology on A induced by giving every A_α the subspace topology agrees with the subspace topology of A as a subspace of $\prod_\alpha X_\alpha$ with the product topology.

Some really bizarre things can happen when we take arbitrary products. Recall that a topological space is Fréchet-Urysohn if for every subset $S \subseteq X$ and every $s \in \overline{S}$, there exists a sequence (s_n) of points in S converging to s. We are now ready to construct a space that does not have this property: $\mathbb{R}^J$, where J is any uncountable set and we endow $\mathbb{R}$ with the standard topology. Note that this is a Hausdorff space, since it is the product of Hausdorff spaces (using Exercise 1.11.14).

Example 1.11.15. Let J be an uncountable set (for example, $J = \mathbb{R}$), and consider the product $\mathbb{R}^J$. Endow $\mathbb{R}$ with the standard topology and $\mathbb{R}^J$ with the product topology. We get a basis for the topology on $\mathbb{R}^J$ when we take the set of all

$$U = \prod_{\alpha \in J} U_\alpha$$

where we select a finite number of $\alpha \in J$, say $\alpha_1, \dots, \alpha_n$, for each $= 1, \dots, n$, and we pick an open interval $U_{\alpha_i} = (a_i, b_i)$. Viewing $\mathbb{R}^J$ as the set of all functions from J to $\mathbb{R}$, we can describe our basis as the set of all

$$U = \left\{ f \colon J \longrightarrow \mathbb{R} \mid f(\alpha_i) \in (a_{\alpha_i}, b_{\alpha_i}) \text{ for all } i = 1, \dots, n \right\}.$$

Thus our basis is the collection of functions that must pass through a finite number of specified windows.

Let

$$S := \{ f \colon J \longrightarrow \mathbb{R} \mid f(x) = 1 \text{ for all but finitely many values of } x \in J \}.$$

Let z denote the constant function equal to 0. We claim that $z \in \overline{S}$. Fix any basic open neighborhood U of 0 in $\mathbb{R}^J$, where

$$U = \prod_{\alpha \in J} U_\alpha \qquad \text{with } U_\alpha = \begin{cases} (a_i, b_i) & \text{if } \alpha = \alpha_i \\ \vdots & \\ \mathbb{R} & \text{if } \alpha \notin \{\alpha_1, \dots, \alpha_n\}. \end{cases}$$

Since $0 \in U$, we have $a_i < 0 < b_i$ for all i. By setting

$$f(\alpha) = \begin{cases} 0 & \text{if } \alpha = \alpha_1 \\ \vdots & \\ 0 & \text{if } \alpha = \alpha_n \\ 1 & \text{if } \alpha \notin \{\alpha_1, \dots, \alpha_n\} \end{cases}$$

we see that

$$f = (f(\alpha))_{\alpha \in J} \in U \cap S.$$

This proves that $z \in \overline{S}$.

However, we claim that no sequence of elements of S converges to z. Suppose that $(f_n)_{n \geqslant 1}$ is any sequence of elements in S that converges to some element $g \in \mathbb{R}^J$. For each n, let

$$T_n = \{ x \in J \mid f_n(x) \neq 1 \}.$$

By definition of S, T_n is a finite set, so

$$T := \bigcup_n T_n$$

is countable since, more generally, any countable union of countable sets is countable. In particular, $T \neq J$. Let $x \in J \setminus T$. Since (f_n) converges to g, then $(f_n(x))$ converges to $g(x) \in \mathbb{R}$. By definition of T, $f_n(x) = 1$ for all n, and thus $g(x) = 1$. In particular, $g \neq z$.

1.12 Metric spaces

Definition 1.12.1. A **metric** on a set X is a function $d\colon X \times X \longrightarrow \mathbb{R}$ satisfying the following properties:

- For all $x, y \in X$, $d(x, y) \geqslant 0$, and equality holds if and only if $x = y$.
- For all $x, y \in X$, $d(x, y) = d(y, x)$.
- Triangle inequality: for all $x, y, z \in X$,

$$d(x, y) + d(y, z) \geqslant d(x, z)$$

Fix a metric d on X, a point $x \in X$, and a real number $\varepsilon > 0$. The **open ball** of radius ε centered at x relative to d, also sometimes call the **ε-ball centered at** x, is given by

$$B_\varepsilon^d(x) := \{y \in X \mid d(x, y) < \varepsilon\}.$$

Example 1.12.2. The **Euclidean metric** on $\mathbb{R}^n$ is given by

$$d((x_1, \dots, x_n), (y_1, \dots, y_n)) = \sqrt{\sum_i (x_i - y_i)^2}.$$

When $n = 2$, $B_\varepsilon^d(x, y)$ is a the open disc of radius ε centered at (x, y).

Example 1.12.3. The **square metric** on $\mathbb{R}^n$ is given by

$$\rho((x_1, \dots, x_n), (y_1, \dots, y_n)) = \max\{|x_i - y_i| \mid i = 1, \dots n\}.$$

When $n = 2$, $B_\varepsilon^\rho(x, y)$ is a the open square of side length 2ε centered at (x, y).

Example 1.12.4. The **taxi-cab metric** is given by

$$d^{\text{taxi}}((x_1, \dots, x_n), (y_1, \dots, y_n)) = \sum_i |x_i - y_i|.$$

When $n = 2$, $B_\varepsilon^{d^{\text{taxi}}}(x, y)$ is a the open diamond of radius ε centered at (x, y).

Lemma 1.12.5. *Let d be a metric a set X. The set*

$$B^d(X) := \{B_\varepsilon^d(x) \mid x \in X \text{ and } \varepsilon > 0\}$$

is a basis for a topology on X, called the **metric topology** *associated to d. Moreover, the set of open balls of radius $\varepsilon < M$ for a fixed M is also a basis for the topology on X.*

Proof. For each $x \in X$, the open ball $B_\varepsilon(x)$ contains x for any $\varepsilon > 0$; in particular, we can take any $\varepsilon < M$. Now before we continue on to proving we have a basis, we need an auxiliary fact first: that if $y \in \mathscr{B}_\varepsilon(x)$, then there exists $\delta > 0$ such that $B_\delta(y) \subseteq B_\varepsilon(x)$. To do that, first note that by definition of $B_\varepsilon(x)$, we have $d(x, y) < \varepsilon$, and thus

$$\delta := \varepsilon - d(x, y) > 0.$$

Now if we take any $z \in B_\varepsilon(y)$, by the Triangle Inequality

$$d(x,z) < d(x,y) + d(y,z) < d(x,y) + \delta = d(x,y) + (\varepsilon - d(x,y)) = \varepsilon.$$

Therefore, $B_\delta(y) \subseteq B_\varepsilon(x)$.

Now consider any two open balls $B_\varepsilon(x)$ and $B_\delta(y)$ be two open balls with $z \in B_\varepsilon(x) \cap B_\delta(y)$. Then there exists an open ball $B_{v_x}(z) \subseteq B_\varepsilon(x)$ and an open ball $B_{v_y}(z) \subseteq B_\delta(y)$. Take $v := \min\{v_x, v_y\} > 0$. Then

$$B_v(z) \subseteq B_{v_x}(x) \quad \text{and} \quad B_v(z) \subseteq B_{v_y}(y), \text{ so } B_v(z) \subseteq B_\varepsilon(x) \cap B_\delta(y).$$

This shows that $B^d(X)$ is indeed a basis for X. Moreover, if we require $\varepsilon, \delta < M$, note that $v \leqslant v_x \leqslant \varepsilon < M$, and thus the set of open balls of radius smaller than M is still a basis for the same topology. □

Remark 1.12.6. Let d be a metric on a set X. A subset $U \subseteq X$ is open for the metric topology if and only if for every $a \in U$ there exists $\varepsilon > 0$ such that $B^d_\varepsilon(a) \subseteq U$.

Example 1.12.7. The metric topology on $\mathbb{R}^n$ associated to the Euclidean metric is the standard Euclidean topology.

Example 1.12.8. Let X be any set, and let

$$d(x,y) = \begin{cases} 1 & \text{if } x \neq y \\ 0 & \text{if } x = y. \end{cases}$$

One can check that this is indeed a metric. In this metric,

$$\{x\} = B_{\frac{1}{2}}(x) \quad \text{for all } x \in X.$$

Thus the topology induced by this metric is the discrete topology.

Theorem 1.12.9. *Let X be a set and d_1 and d_2 metrics on X. The topology $\mathcal{T}_1$ associated to d_1 is finer than the topology $\mathcal{T}_2$ associated to d_2 if and only if for every $a \in X$ and $\varepsilon > 0$ there exists $\delta > 0$ such that*

$$B^{d_1}_\delta(a) \subseteq B^{d_2}_\varepsilon(a).$$

Proof. ($\Leftarrow$): Let $U \in \mathcal{T}_2$ and $a \in U$. By Remark 1.12.6, there is a $\varepsilon > 0$ such that $B^{d_2}_\varepsilon(a) \subseteq U$ and thus, by assumption, there is a $\delta > 0$ such that $a \in B^{d_1}_\delta(a) \subseteq U$. This proves U is open in $\mathcal{T}_1$.

($\Rightarrow$): Choose $a \in X$ and $\varepsilon > 0$. Then $U := B^{d_2}_\varepsilon(a)$ belongs to $\mathcal{T}_2$ and so, by assumption, U belongs to $\mathcal{T}_1$ too. By Remark 1.12.6, given a point $a \in U$, there exists $\delta > 0$ such that $B^{d_1}_\delta(a) \subseteq U = B^{d_2}_\varepsilon(a)$. □

Definition 1.12.10. Let (X,d) be a metric space. A subset $A \subseteq X$ is **bounded** if there exists $M > 0$ such that

$$d(x,y) \leqslant M \qquad \text{for all } x, y \in A.$$

If A is bounded and nonempty, its **diameter** is

$$\sup\{d(a,b) \mid a, b, \in A\}.$$

Boundedness is not a topological invariant.

Lemma 1.12.11. *Given any metric d on a set X, define $\overline{d}$ by the rule*

$$\overline{d}(x,y) = \min\{d(x,y), 1\}.$$

Then $\overline{d}$ is a metric that is topologically equivalent to d.

Proof. Given any $x, y \in X$, either $\overline{d}(x,y) = 1 \geqslant 0$ or $\overline{d}(x,y) = d(x,y) \geqslant 0$ since d is a metric. Moreover, also because d is a metric,

$$\overline{d}(x,y) = 0 \iff d(x,y) = 0 \iff x = y.$$

Now let us check the Triangle Inequality holds for $\overline{d}$. First, note that $d(x,y) \geqslant \overline{d}(x,y)$ for all $x, y \in X$. Given $x, y, z \in X$, if $d(x,y) \geqslant 1$ or $d(y,z) \geqslant 1$, then $\overline{d}(x,y) = 1$ or $\overline{d}(y,z) = 1$, so

$$\overline{d}(x,y) + \overline{d}(y,z) \geqslant 1 \geqslant \overline{d}(x,z).$$

If both $d(x,y), d(y,z) \leqslant 1$, then

$$\overline{d}(x,y) + \overline{d}(y,z) = d(x,y) + d(y,z) \geqslant d(x,z) \geqslant \overline{d}(x,z).$$

This shows the triangle inequality holds, and thus $\overline{d}$ is a metric.

Now it remains to see that $\overline{d}$ is topologically equivalent to d. By Lemma 1.12.5, the set of all open balls of radius less than 1 forms a basis for the topology induced by d and the topology induced by $\overline{d}$. But for any $\varepsilon < 1$,

$$d(x,y) < \varepsilon \iff \overline{d}(x,y) < \varepsilon$$

so $B_\varepsilon^d(x) = B_\varepsilon^{\overline{d}}(x)$. We have thus provided the *same* basis for both topologies, and we conclude that d and $\overline{d}$ are topologically equivalent metrics. □

Remark 1.12.12. If (X,d) is a metric space for which there exists an unbounded set A, then A becomes bounded under the metric $\overline{d}$ in Lemma 1.12.11, which induces the same topology as d. This shows that the property of being bounded is not a topological invariant.

Two different metrics can determine the same topology!

Definition 1.12.13. We say two metrics d_1 and d_2 on the same set X are **topologically equivalent metrics** if the topologies they generate coincide.

Exercise 1.12.14. Suppose d_1 and d_2 are two metrics on a set X. Show that if there exist strictly positive real numbers r and s such that

$$r \cdot d_1(x,y) \leqslant d_2(x,y) \leqslant s \cdot d_1(x,y)$$

for all $x, y \in X$, then d_1 and d_2 are topologically equivalent.

Example 1.12.15. Consider the Euclidean and square metrics on $\mathbb{R}^n$, which are given by

$$d((x_1, \dots, x_n), (y_1, \dots, y_n)) = \sqrt{\sum_i (x_i - y_i)^2}$$

and

$$\rho((x_1, \dots, x_n), (y_1, \dots, y_n)) = \max\{|x_i - y_i| \mid i = 1, \dots n\}.$$

Since

$$\rho(a, b) \leqslant d(a, b) \leqslant \sqrt{n}\rho(a, b)$$

for all $a, b \in \mathbb{R}^n$, by Exercise 1.12.14 we conclude that these two metrics are equivalent, and thus both give us the same topology on $\mathbb{R}^n$: the standard topology.

Definition 1.12.16. A topological space $(X, \mathcal{T})$ is **metrizable** if there exists a metric on X that induces the topology $\mathcal{T}$. A **metric space** (X, d) is a metrizable space X together with a metric that induces the topology on X.

As we have seen in Example 1.12.15, a metric d that induces the topology on the metrizable space X is typically *not* unique.

Example 1.12.17. Example 1.12.15 shows us two metrics inducing the standard topology on $\mathbb{R}^n$, and thus $\mathbb{R}^n$ with the standard topology is a metrizable space.

Example 1.12.18. By Example 1.12.8, any space with the discrete topology is metrizable.

Not every topological space is metrizable. In fact, metrizable spaces have nice properties.

Theorem 1.12.19. *Every metrizable topological space is Hausdorff.*

Proof. Let d be a metric on X that generates the topology. For $x, y \in X$ with $x \neq y$, set $\varepsilon = \frac{1}{2}d(x, y)$. Then $B_\varepsilon^d(x)$ and $B_\varepsilon^d(y)$ are open neighborhoods of a and b that do not meet: if $z \in B_\varepsilon^d(x)$, then by the Triangle Inequality

$$d(x, z) + d(z, y) \geqslant d(x, y) = \varepsilon \implies d(z, y) \geqslant \varepsilon - d(x, z) = \varepsilon - \frac{\varepsilon}{2} = \frac{\varepsilon}{2}$$

and thus $z \notin B_\varepsilon^d(y)$. □

Example 1.12.20. The line with two origins is not metrizable, since it is not Hausdorff.

Theorem 1.12.21. *Every metrizable space X is Fréchet-Urysohn: for every subset $S \subseteq X$ and every point $a \in \overline{S}$, there is a sequence $(s_n)_{n \geqslant 1}$ in S converging to a.*

Proof. Given $a \in \overline{S}$, for each integer $n \geqslant 1$, the ball $B_{\frac{1}{n}}^d(a)$ is an open neighborhood of a, and thus it must meet S. Let $s_n \in S \cap B_{\frac{1}{n}}^d(a)$. We claim that $(s_n)_{n \geqslant 1}$ converges to a. By Remark 1.12.6, given any open neighborhood U of a, there exists $\varepsilon > 0$ such that $B_\varepsilon^d(a) \subseteq U$. For all $n \geqslant \frac{1}{\varepsilon}$,

$$d(a, s_n) < \frac{1}{n} \leqslant \varepsilon \implies s_n \in B_\varepsilon^d(a) \subseteq U.$$

We conclude that (s_n) converges to a. □

Remark 1.12.22. We saw in Exercise 1.9.28 that every continuous function sends convergent sequences to convergent sequences, and that the converse holds when the domain is Fréchet-Urysohn. Given Theorem 1.12.21, we can now say that if $f\colon X \longrightarrow Y$ is a function with metrizable domain X, then

$$f \text{ is continuous } \iff \text{ if } (x_n)_{n\geqslant 1} \text{ converges to } x \text{ in } X, \text{ then } (f(x_n))_{n\geqslant 1} \text{ converges to } y \text{ in } Y.$$

Example 1.12.23. The product $\mathbb{R}^{\mathbb{R}}$ with the product topology induced by the standard topology on $\mathbb{R}$ is not metrizable, since it is not Fréchet-Urysohn by Example 1.11.15.

We will soon see that in contrast, $\mathbb{R}^{\omega}$ with the product topology (and the standard topology on each copy of $\mathbb{R}$) is metrizable. First, here is another metric on any product of copies of $\mathbb{R}$.

Definition 1.12.24. Let $\overline{d}$ be the standard bounded metric on $\mathbb{R}$, meaning that

$$d(x,y) = \min\{|x-y|, 1\}.$$

Given any set I, the **uniform metric** on $\mathbb{R}^I$ is the metric determined as follows: given $x = (x_\alpha)_{\alpha\in I}$ and $y = (y_\alpha)_{\alpha\in I}$

$$\overline{\rho}(x,y) = \sup\{\overline{d}(x_\alpha, y_\alpha) \mid \alpha \in I\}.$$

The topology induced by $\overline{\rho}$ is called the **uniform topology** on $\mathbb{R}^I$.

Exercise 1.12.25. Check that the uniform metric on $\mathbb{R}^I$ is indeed a metric for any set I.

Theorem 1.12.26. *Let I be any set and consider the Tychonoff, uniform, and box topologies on $\mathbb{R}^I$, with the Tychonoff and box topologies induced by the standard topology on $\mathbb{R}$, which we will denote respectively by $\mathcal{T}_{\textit{Tychonoff}}$, $\mathcal{T}_{\textit{uniform}}$, and $\mathcal{T}_{\textit{box}}$. Then*

$$\mathcal{T}_{\textit{Tychonoff}} \subseteq \mathcal{T}_{\textit{uniform}} \subseteq \mathcal{T}_{\textit{box}}.$$

Proof. Fix a nonempty basic open set

$$U = \prod_{\alpha\in I} U_\alpha$$

for the product topology, so that U_α is open in the standard topology on $\mathbb{R}$ and $U_\alpha = \mathbb{R}$ for all but finitely many $\alpha \in I$, say $\alpha_1, \dots, \alpha_n$. Consider a point $x = (x_\alpha)_{\alpha\in I}$ in U.

For each of $\alpha_1, \dots, \alpha_n$, fix ε_i such that

$$0 < \varepsilon_i < 1 \qquad \text{and} \qquad (x_{\alpha_i} - \varepsilon_i, x_{\alpha_i} + \varepsilon_i) \subseteq U_{\alpha_i}.$$

Set

$$\varepsilon := \min_{1\leqslant i\leqslant n} \varepsilon_i.$$

Note that $1 > \varepsilon > 0$, and that for all $\alpha \in I$,

$$(x_\alpha - \varepsilon, x_\alpha + \varepsilon) \subseteq U_\alpha.$$

When $\alpha \notin \{\alpha_1, \dots, \alpha_n\}$, this is a triviality, and for $\alpha_1, \dots, \alpha_n$ this is a consequence of the definition of ε. Then

$$B_\varepsilon^{\overline{\rho}}(x) \subseteq U.$$

Thus U is open in the uniform topology.

Now fix an open ball $B_\varepsilon^{\overline{\rho}}(x)$ with $\varepsilon < 1$; the set of all such balls forms a basis for the uniform topology. In the proof of Lemma 1.12.5, we showed that for all $y \in B_\varepsilon^{\overline{\rho}}(x)$, there exists δ such that $B_\delta^{\overline{\rho}}(y) \subseteq B_\varepsilon^{\overline{\rho}}(x)$. Thus after replacing x with y and ε with δ, it suffices to show that there exists an open set U in the box topology such that $x \in U \subseteq B_\varepsilon^{\overline{\rho}}(x)$, and that will prove that $B_\varepsilon^{\overline{\rho}}(x)$ is open in the box topology. Set

$$U = \prod_{\alpha \in I} (x_\alpha - \frac{\varepsilon}{2}, x_\alpha + \frac{\varepsilon}{2}),$$

which is open in the box topology. Then for any $y \in U$ and any $\alpha \in I$,

$$\overline{d}(x_\alpha, y_\alpha) < \frac{\varepsilon}{2} < 1 \implies d(x_\alpha, y_\alpha) = \overline{d}(x_\alpha, y_\alpha) < \frac{\varepsilon}{2}$$

and thus

$$\overline{\rho}(x, y) = \sup\left\{\overline{d}(x_\alpha, y_\alpha) \mid \alpha \in I\right\} \leqslant \frac{e}{2} < \varepsilon.$$

Therefore, $y \in B_\varepsilon^{\overline{\rho}}(x)$ and $U \subseteq B_\varepsilon^{\overline{\rho}}(x)$, so $B_\varepsilon^{\overline{\rho}}(x)$ is open in the box topology. □

Remark 1.12.27. Endow $\mathbb{R}$ with the standard topology. Fix an integer $n \geqslant 1$. By Exercise 1.4.7, the product and box topologies on $\mathbb{R}^n$ coincide with the standard topology, and thus by Theorem 1.12.26 the uniform topology on $\mathbb{R}^n$ is the standard topology.

Moreover, given $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$, set

$$d(x, y) := \max_{1 \leqslant i \leqslant n} \{|x_i - y_i|\},$$

the taxicab metric on $\mathbb{R}^n$. By Lemma 1.12.11, d is topologically equivalent to the uniform metric on $\mathbb{R}^n$. In particular, this shows that the taxicab metric induces the standard topology on $\mathbb{R}^n$.

Exercise 1.12.28. Endow $\mathbb{R}$ with the standard topology. Show that if I is any infinite set, then the Tychonoff, uniform, and box topologies on $\mathbb{R}^I$ are all distinct.

Nevertheless, the product topology on $\mathbb{R}^\omega$ is metrizable.

Definition 1.12.29. Consider the standard bounded metric on $\mathbb{R}$,

$$d(x, y) = \min\{|x - y|, 1\}.$$

Given $x = (x_n)_{n \geqslant 1}, y = (y_n)_{n \geqslant 1} \in \mathbb{R}^\omega$, set

$$D(x, y) := \sup_i \left\{\frac{\overline{d}(x_i, y_i)}{i}\right\}.$$

Exercise 1.12.30. Show that D is a metric on $\mathbb{R}^\omega$.

We will now prove that this metric induces the product topology on $\mathbb{R}^\omega$, showing that $\mathbb{R}^\omega$ is metrizable.

Theorem 1.12.31. *The metric D induces the product topology on $\mathbb{R}^\omega$.*

Proof. First, one needs to check that D is indeed a metric on $\mathbb{R}^\omega$; we leave the details for Exercise 1.12.30. To show that D induces the product topology, first consider an open ball $B_\varepsilon^D(x)$ in the metric topology induced by D. Choose N large enough so that $\frac{1}{N} < \varepsilon$, and let

$$V = (x_1 - \varepsilon, x_1 + \varepsilon) \times \cdots \times (x_N - \varepsilon, x_N + \varepsilon) \times \prod_{n=N+1}^{\infty} \mathbb{R},$$

which is open in the product topology. Given any $i \geqslant N$ and any $y_i \in \mathbb{R}$, note that

$$\overline{d}(x_i, y_i) \leqslant \text{ and } \frac{1}{i} \leqslant \frac{1}{N} \implies \frac{\overline{d}(x_i, y_i)}{i} \leqslant \frac{1}{N}.$$

Therefore, given any $y \in \mathbb{R}^\omega$,

$$D(x, y) \leqslant \max\left\{ \frac{\overline{d}(x_1, y_1)}{1}, \ldots, \frac{\overline{d}(x_N, y_N)}{N}, \frac{1}{N} \right\}.$$

In particular, if $y \in V$, then

$$\frac{\overline{d}(x_i, y_i)}{i} < \varepsilon$$

for all $i \leqslant N$, and since we chose N so that $\frac{1}{N} < \varepsilon$, we have

$$D(x, y) < \varepsilon.$$

We conclude that $x \in V \subseteq B_\varepsilon^D(x)$.

Conversely, given an open set in the product topology

$$U = \prod_{i=1}^{n} U_i,$$

so that U_i is open in $\mathbb{R}$ and $U_i = \mathbb{R}$ for all $i \geqslant N$ for some N, fix $x \in U$. For each $i \leqslant N$, let $\varepsilon_i > 0$ be such that

$$(x - \varepsilon_i, x + \varepsilon_i) \subseteq U_i$$

and set

$$\varepsilon = \min\left\{ \frac{\varepsilon_i}{i} \mid 1 \leqslant i \leqslant N \right\}.$$

Then for all $y \in B_\varepsilon^D(x)$ and $1 \leqslant i \leqslant N$, we have

$$\frac{\overline{d}(x_i, y_i)}{i} \leqslant D(x, y) < \varepsilon \leqslant \frac{\varepsilon_i}{i},$$

so in particular $y_i \in (x - \varepsilon_i, x + \varepsilon_i) \subseteq U_i$ and $y \in U$. We conclude that $x \in B_\varepsilon^D(x) \subseteq U$. $\square$

The uniform topology, however, has the added advantage that it allows us to talk about uniform convergence, a stronger form of convergence.

Remark 1.12.32. Fix a set J and a topological space X. Thinking of X^J as the set of functions $J \longrightarrow X$, a sequence $(f_n)_{n\geqslant 1}$ of functions $f_n \colon J \longrightarrow X$ can be reinterpreted as a sequence of elements in X^J.

Definition 1.12.33. Let J be a set and (X, d) be a metric space. A sequence $(f_n)_{n\geqslant 1}$ of functions $f_n \colon J \longrightarrow X$ **converges uniformly** to $f \colon J \longrightarrow X$ if

$$\text{for all } \varepsilon > 0 \text{ there exists } N \geqslant 1 \text{ such that for all } n \geqslant N \text{ and } \alpha \in J, \quad d(f_n(\alpha), f(\alpha)) < \varepsilon.$$

Exercise 1.12.34. Endow $\mathbb{R}$ with the standard topology. Show that a sequence $(f_n)_{n\geqslant 1}$ of functions $f_n \colon J \longrightarrow \mathbb{R}$ converges uniformly to $f \colon J \longrightarrow \mathbb{R}$ if and only if the sequence $(f_n)_{n\geqslant 1}$ of points in $\mathbb{R}^J$ converges to f in the uniform topology.

Theorem 1.12.35 (Uniform Limit Theorem)**.** *Let X be a topological space, (Y, d) a metric space, and $(f_n)_{n\geqslant 1}$ a sequence of continuous functions $f_n \colon X \longrightarrow Y$. If (f_n) converges to $f \colon X \longrightarrow Y$ uniformly, then f is continuous.*

Proof. Let V be an open set in Y, and fix $x \in f^{-1}(V)$. Since V is open in the metric topology, there exists $\varepsilon > 0$ such that $B_\varepsilon(f(x)) \subseteq V$. Since (f_n) converges uniformly to f, there exists N such that for all $n \geqslant N$ and all $z \in X$,

$$d(f_n(z), f(z)) < \frac{\varepsilon}{3}.$$

Since f_N is continuous itself, and $B_{\frac{\varepsilon}{3}}(f_N(x))$ is open in Y, there exists an open neighborhood U of x such that

$$U \subseteq f_N^{-1}(B_{\frac{\varepsilon}{3}}(f_N(x))),$$

or equivalently, $f_N(U) \subseteq B_{\frac{\varepsilon}{3}}(f_N(x))$, which we can rewrite as

$$d(f_N(z), f_N(x)) < \frac{\varepsilon}{3} \text{ for all } z \in U.$$

We claim that $U \subseteq f^{-1}(V)$. Given $z \in U$, we chose N so that

$$d(f_N(z), f(z)) < \frac{\varepsilon}{3} \quad \text{and} \quad d(f_N(x), f(x)) < \frac{\varepsilon}{3}$$

and we chose U so that

$$d(f_N(x), f_N(x)) < \frac{e}{3}.$$

By the Triangle Inequality, we see that for all $z \in U$,

$$\begin{aligned} d(f(z), f(x)) &\leqslant d(f(z), f_N(z)) + d(f_N(z), f(x)) \\ &\leqslant d(f(z), f_N(z)) + d(f_N(z), f_N(x)) + d(f_N(x), f(x)) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Therefore, $f(z) \in B_\varepsilon(f(x))$ for all $z \in U$, so $U \subseteq f^{-1}(B_\varepsilon(f(x))$ and f is continuous. $\square$

1.13 Quotient spaces

We can construct topological spaces from others by *glueing* together points. Many spaces of interest arise in this way. First, we will carefully introduce the abstract idea, and then see many examples.

The key idea is that any surjective map from a topological space induces a natural topology on the codomain.

Lemma 1.13.1. *Let X be a topological space and $p\colon X \twoheadrightarrow Y$ a surjective function. Then*

$$\mathcal{T} = \{U \subseteq Y \mid p^{-1}(U) \text{ is open in } X\}$$

is a topology on Y.

Proof. Since

$$p^{-1}(\emptyset) = \emptyset \quad \text{and} \quad p^{-1}(Y) = X$$

and X and $\emptyset$ are open in X, then $\emptyset, Y \in \mathcal{T}$.

Let $U, V \in \mathcal{T}$. By Exercise 1.9.1,

$$p^{-1}(U \cap V) = p^{-1}(U) \cap p^{-1}(V)$$

is an intersection of open sets, and thus open. Thus $U \cap V \in \mathcal{T}$.

Let $\{U_\alpha\}$ be a collection of elements of $\mathcal{T}$. By Exercise 1.9.1,

$$p^{-1}\left(\bigcup_{\alpha\in I} U_\alpha\right) = \bigcup_{\alpha\in I} p^{-1}\left(U_\alpha\right)$$

is a union of open sets, and thus open. We conclude that

$$\bigcup_{\alpha\in I} U_\alpha \in \mathcal{T}.$$

□

Definition 1.13.2. Let X is a topological space and $p\colon X \twoheadrightarrow Y$ is a surjective function. The **quotient topology** on Y (with respect to p) is the topology

$$T = \{U \subseteq Y \mid p^{-1}(U) \text{ is open in } X\}.$$

Remark 1.13.3. Given a topological space X and any equivalence relation $\sim$ on the set X, the canonical map $p\colon X \longrightarrow X/\sim$ sending any element to its equivalence class $x \mapsto [x]$ is a surjection. By Lemma 1.13.1, p induces a topology on the set $X/\sim$ of equivalence classes, where $U \subseteq X/\sim$ is open if and only if

$$\{x \in X \mid [x] \in U\}$$

is open in X. We call this the **quotient topology** on $X/\sim$. In particular, open sets in the quotient space $X/\sim$ correspond to unions of equivalence classes in X.

We can reframe this in terms of quotient maps.

Definition 1.13.4. Let X and Y be topological spaces. A surjective function $p\colon X \longrightarrow Y$ is a **quotient map** if for all subsets $U \subseteq Y$,

$$U \text{ is open in } Y \iff p^{-1}(U) \text{ is open in } X.$$

Remark 1.13.5. Any quotient map is in particular continuous. However, this condition is stronger than continuity, and it is in fact sometimes called *strong continuity*.

Remark 1.13.6. Given a topological space X and a surjective map $p\colon X \longrightarrow Y$, the quotient topology on Y is the unique topology that makes p a quotient map.

Remark 1.13.7. Let X be a topological space, let $\sim$ be an equivalence relation on X, and consider the canonical map $p\colon X \longrightarrow X/\sim$. Then p is a quotient map when we endow $X/\sim$ with the quotient topology.

Exercise 1.13.8. Given a topological space X and a surjective map $p\colon X \longrightarrow Y$, show that the quotient topology on Y is the finest topology such that p is continuous.

We can also describe a quotient map in terms of saturated sets.

Definition 1.13.9. Let $p\colon X \longrightarrow Y$ be a surjective function. A subset C of X is **saturated** with respect to p if for all $y \in Y$,

$$p^{-1}(\{y\}) \cap C \neq \emptyset \iff p^{-1}(\{y\}) \subseteq C.$$

Remark 1.13.10. A subset $C \subseteq X$ is saturated with respect to $p\colon X \longrightarrow Y$ if only if it is a union of fibers for p. Equivalently, C is saturated if and only if $C = p^{-1}(U)$ for some $U \subseteq Y$. Thus a surjective map $p\colon X \longrightarrow Y$ between two topological spaces is a quotient map if and only if p is continuous and it sends saturated open sets to open sets.

Remark 1.13.11. Suppose that $p\colon X \longrightarrow Y$ is a surjective continuous map between two topological spaces. Note that p is a quotient map if and only if for all subsets $U \subseteq Y$,

$$p^{-1}(U) \text{ open in } X \implies U \text{ open in } Y.$$

Lemma 1.13.12. *Let $p\colon X \longrightarrow Y$ be a surjective continuous map between topological spaces. If p is open or closed, then p is a quotient map.*

Proof. If p is an open map, then in particular p sends saturated open sets to open sets. Since p is surjective, for all $U \subseteq Y$ we have $U = p(p^{-1}(U))$. If $p^{-1}(U)$ is open, then we must have U open. Therefore, p is a quotient map.

Similarly, suppose that p is closed. Then for all $U \subseteq Y$ such that $p^{-1}(U)$ is open, then

$$p^{-1}(Y \setminus U) = X \setminus p^{-1}(U)$$

is closed. Since p is surjective,

$$Y \setminus U = p(p^{-1}(Y \setminus U))$$

and this is the image of a closed set, and thus closed. We conclude that p is a quotient map. □

However, not all quotient maps are open or closed.

Example 1.13.13. Endow $\mathbb{R}$ with the standard topology and consider the set $Y = (-\infty, 0) \cup \{q\} \cup [1, \infty)$. Let $p \colon \mathbb{R} \longrightarrow Y$ be the map given by

$$p(x) = \begin{cases} x & \text{if } x < 0 \text{ or } x \geqslant 1 \\ p & \text{if } 0 \leqslant x < 1. \end{cases}$$

By construction, p is surjective, so we can endow Y with the quotient topology induced by p. In fact, Y is the space obtained from $\mathbb{R}$ by identifying all points in $[0, 1)$ as the same point q, and p is the canonical quotient map.

We claim that p is neither open nor closed. It is not open because the open set $U = (0, 1)$ in $\mathbb{R}$ is taken to $p(U) = \{q\}$, which is not open in Y since

$$p^{-1}(\{q\}) = [0, 1)$$

is not open in $\mathbb{R}$. It is not closed because the closed set $C = \{0\}$ in $\mathbb{R}$ is taken to $p(C) = \{q\}$ in Y, which is not closed since

$$p^{-1}(\{q\}) = [0, 1)$$

is not closed in $\mathbb{R}$.

Lemma 1.13.14. *Let $p \colon X \longrightarrow Y$ be a quotient map between topological spaces, and $\sim$ the equivalence relation on X induced by p, so that $x \sim y$ exactly when $p(x) = p(y)$. Then Y is homeomorphic to $X/\sim$ under the quotient topology.*

Proof. Define $g \colon Y \longrightarrow X/\sim$ as follows: given $y \in Y$, since p is surjective we can find $x \in X$ such that $p(x) = y$, and we set

$$g(y) = [x].$$

This map g is well-defined: given any other $z \in X$ such that $p(z) = y = p(x)$, by definition $z \sim x$, and thus $[z] = [x]$. Moreover, g is injective by construction: given $y_1 \neq y_2$ in Y, any $x_1, x_2 \in X$ with $p(x_1) = y_1$ and $p(x_2) = y_2$ necessarily have $p(x_1) \neq p(x_2)$, so $[x_1] \neq [x_2]$, and thus $g(y_1) \neq g(y_2)$. Finally, g is surjective: given any class $[x] \in X/\sim$, and any representative x of $[x]$,

$$[x] = g(p(x)).$$

Thus g is a bijective function.

We claim that g is in fact a homeomorphism. For any subset U of Y,

$$g(U) = \{[x] \mid p(x) \in U\} = \{[x] \mid x \in p^{-1}(U)\} = q(p^{-1}(U)).$$

Since g is surjective, $g^{-1}(g(U)) = U$. Then

$$\begin{aligned} U \text{ is open in } Y &\iff p^{-1}(U) \text{ is open in } X && \text{since } p \text{ is a quotient map} \\ &\iff q(p^{-1}(U)) \text{ is open in } X/\sim && \text{by definition of the quotient topology} \\ &\iff g(U) \text{ is open in } X/\sim . \quad \square \end{aligned}$$

Definition 1.13.15. Let X and Y be topological spaces. We say Y is a **quotient** of X if there exists a quotient map $p\colon X \longrightarrow Y$.

Lemma 1.13.14 says that a quotient space of X is precisely one that is obtained (up to homeomorphism) via an equivalence relation on X.

Theorem 1.13.16 (Universal Property for Quotient Maps)**.** *Let $p\colon X \longrightarrow Y$ be a quotient map between topological spaces. Given any topological space Z and a function $f\colon X \longrightarrow Z$ that is constant on all fibers, that is, such that*

$$p(a) = p(b) \implies f(a) = f(b),$$

there exists a unique function $\overline{f}\colon Y \longrightarrow Z$ such that $\overline{f} \circ p = f$, so that the following diagram commutes:

$$\begin{array}{ccc} X & \xrightarrow{p} & Y \\ \downarrow{\scriptstyle f} & \swarrow_{f} & \\ Z & & \end{array}$$

Moreover, if $\overline{f}$ is continuous if and only if f is continuous, and $\overline{f}$ is a quotient map if and only if f is a quotient map.

Proof. First, let us define $\overline{f}$. Given $y \in Y$, take any $x \in X$ such that $p(x) = y$, which exists since p is surjective, and set

$$\overline{f}(y) = f(x).$$

This is well-defined: if we take any other z such that $p(z) = y = p(x)$, then by assumption $f(z) = f(x)$. Moreover, this function $\overline{f}$ does in fact make the diagram commute: for each $x \in X$,

$$\overline{f} \circ p(x) = f(x)$$

by construction of $\overline{f}$. Finally, note that $\overline{f}$ is the unique function making the diagram commute: given any other g such that $g \circ p = f$ and any $y \in Y$, surjectivity of p gives us $x \in X$ with $p(x) = y$, and

$$g(y) = g \circ p(x) = f(x) = \overline{f}(y).$$

If $\overline{f}$ is continuous, then $f = \overline{f} \circ p$ is a composition of continuous functions, and thus continuous by Lemma 1.9.19. Conversely, if f is continuous, then given any open set U of Z

$$f^{-1}(U) = p^{-1}(\overline{f}^{-1}(U))$$

is open in Y. Since p is a quotient map, $\overline{f}^{-1}(U)$ must be open in Y. We conclude that $\overline{f}$ is also open.

Suppose $\overline{f}$ is a quotient map. Then $f = \overline{f} \circ p$ is the composition of two quotient maps, and thus a quotient map by **??**. Conversely, suppose f is a quotient map. Since $f = \overline{f} \circ p$ is surjective, $\overline{f}$ must also be surjective. Given a subset U of Z such that $\overline{f}^{-1}(U)$ is open, we need to show that U is open. Since p is continuous and $\overline{f}^{-1}(U)$ is open,

$$f^{-1}(U) = p^{-1}(\overline{f}^{-1}(U))$$

is open. Since f is a quotient map, U must be open. □

Let us now see some examples of quotient spaces.

Example 1.13.17 (the line with two origins)**.** We can recast the line with two origins as a quotient space. To do that, start with the disjoint union of two copies of $\mathbb{R}$ in the standard topology, $X = \mathbb{R} \sqcup \mathbb{R}$, where we write $\mathbb{R}_1$ and $\mathbb{R}_2$ for the two disjoint copies of $\mathbb{R}$, and give X the topology where $U \subseteq X$ is open if and only if $U \cap \mathbb{R}_1$ and $U \cap \mathbb{R}_2$ are both open. We leave it as an exercise to check that this is in fact a topology on X.

We then consider the equivalence relation that identifies x in $\mathbb{R}_1$ with the same point x in $\mathbb{R}_2$ for all $x \neq 0$. The key point here is that the two origins, $\mathbf{0}_1$ and $\mathbf{0}_2$, are *not* identified. The resulting quotient space is our old friend the line with two origins. If we identify $\mathbb{R}_1$ and $\mathbb{R}_2$ with their images in the quotient, the quotient topology matches our previous description of the open sets was that U is open if $U \cap \mathbb{R}_1$ and $U \cap \mathbb{R}_2$ are open.

Example 1.13.18 (the circle as a quotient space)**.** Endow $\mathbb{R}$ with the standard topology and $X = [0, 1]$ with the subspace topology. Let $\sim$ be the equivalence relation such with equivalence classes $\{0, 1\}$ and $\{x\}$ for each $0 < x < 1$. Equivalently, this is the relation determined by $0 \sim 1$, where only the two points 0 and 1 are glued together.

We claim $Y = X/\sim$ is homeomorphic to S^1, the unit circle in the Euclidean plane (with the subspace topology). To do that, we will construct an explicit homeomorphism: the map $f \colon X \longrightarrow S^1$ given by

$$f(t) = (\cos(2\pi t), \sin(2\pi t)).$$

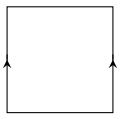
We leave it as an exercise that f is indeed continuous. Note that $f(0) = f(1)$, so f respects the equivalence relation; in other words, f is well-defined. By the universal property for quotient maps, there is an induced continuous map $\overline{f} \colon Y \longrightarrow S^1$. Moreover, using basic trigonometry one can show that $\overline{f}$ is a bijection.

To show that $\overline{f}$ is actually a homeomorphism, it suffices to show that $\overline{f}(U)$ is open in S^1 for all open subsets U of Y. Since Y has the quotient topology, U is open in Y if and only if its preimage in $X = [0, 1]$ under the canonical surjection $p \colon X \longrightarrow Y$ is open. Thus U is open in Y if and only if $p^{-1}(U)$ is a union of open intervals of the form (a, b) with $a, b \in [0, 1]$ and subsets of the form $[0, b) \cup (a, 1]$. One can then check that the images of all of these are open in S^1, and thus $\overline{f}$ is in fact a homeomorphism.

Example 1.13.19 (cylinder)**.** Consider the subspace $X = [0, 1] \times [0, 1]$ of $\mathbb{R}^2$ under the standard topology. The quotient

$$X/\sim = \frac{[0, 1] \times [0, 1]}{(0, y) \sim (1, y)}$$

can be represented by the following picture:



The arrows indicate we will *glue* those two sides to each other, in the same direction. This space is homeomorphic to a cylinder in $\mathbb{R}^3$ under the subspace topology: we leave it as an exercise to check that

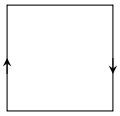
$$(x, y) \mapsto (\cos(2\pi x), \sin(2\pi x), y)$$

determines an embedding $X/\sim = Y \hookrightarrow \mathbb{R}^3$.

Example 1.13.20 (Möbius strip)**.** Again we take a quotient of the rectangle $[0,1] \times [0,1]$ in $\mathbb{R}^2$ under the standard topology; this time,

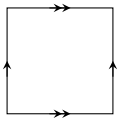
$$\frac{[0,1] \times [0,1]}{(0,y) \sim (1,1-y)}.$$

We glue two opposite sides of the rectangle to each other, but in opposite directions, as in the following diagram:



The resulting space is known as the **Möbius strip**, and can be embedded in $\mathbb{R}^3$.

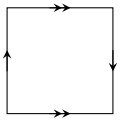
Example 1.13.21 (torus)**.** The **torus** is also an example of a quotient of the rectangle:



Here the different kinds of arrows indicate different segments that need to be glued to each other (glueing only to segments with the same kind of arrow). More formally, the torus is the quotient space

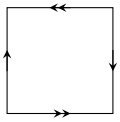
$$\frac{[0,1] \times [0,1]}{(0,y) \sim (1,y), (x,0) \sim (x,1)}.$$

Example 1.13.22 (the Klein Bottle)**.** The **Klein bottle** is the quotient of the rectangle obtained from



This one is much harder to see: in fact, the Klein bottle cannot be embedded in $\mathbb{R}^3$!

Example 1.13.23 (the projective plane)**.** The **projective plane $\mathbf{RP}^2$** is the quotient of the rectangle obtained from



One can show that this space is homeomorphic to the quotient of S^2 obtained by identifying antipodes:

$$\mathbf{RP}^2 = S^2 / \left(x \sim -x \mid x \in S^2 \right).$$

We can embed $\mathbf{RP}^2$ into $\mathbb{R}^4$ as follows: first, one considers the map $\mathbb{R}^3 \longrightarrow \mathbb{R}^4$ given by

$$(x,y,z) \mapsto (xy, xz, y^2 - z^2, 2yz),$$

which one can show is continuous. Restricting to S^2, note that this map takes the same value on (x,y,z) and its antipode $(-x,-y,-z)$, and thus by the universal property of quotients (Theorem 1.13.16) it induces a map from $\mathbf{RP}^2$ to $\mathbb{R}^4$. One then shows it is injective, and in fact a homeomorphism onto into image.

Example 1.13.24 (Real projective space)**.** More generally, $\mathbf{RP}^n$ is the quotient of S^n obtained by identifying antipodes:

$$\mathbf{RP}^n = S^n / \left(x \sim -x \mid x \in S^n \right).$$

Equivalently, we can construct $\mathbf{RP}^n$ as a quotient of $\mathbb{R}^{n+1} \setminus \{0\}$ under the subspace standard topology by identifying $x \neq 0$ with any point λx with $\lambda \neq 0$ any scalar.

Appendix A

Sets and functions

In order to do any topology, we need to be comfortable with elementary set theory. Munkres [Mun00] includes an introduction to this topic; we include here the most important definitions, facts, and notation for completeness, but this is intended as a very quick refresher.

A.1 Sets

Here is some of the notation we will use for sets:

$a \in A$	a is an element of the set A
$a \notin A$	a is not an element of A, which is the negation of $a \in A$
$P(x)$	a statement about the variable x; may be true or false depending on x
$\{x \in D \mid P(x)\}$ also denoted by $\{x \in D \colon P(x)\}$	The set of all elements of D for which $P(x)$ is true (a subset of D)
$A \subseteq B$	A is a **subset** of $B \iff$ If $a \in A$, then $a \in B$.
$A \subsetneq B$	$A \subseteq B$ and $A \neq B$
$A \supseteq B$	B is a subset of A (we also say that A is a superset of B)
$A \supsetneq B$	$A \supseteq B$ and $A \neq B$
$A \not\subseteq B$	A is not a subset of B, which is the negation of $A \subseteq B$
$A \setminus B$	$\{a \in A \mid a \notin B\}$, the complement of A and B, read A *except* B
$A \cap B$	$\{x \mid x \in A$ or $x \in B\}$, the intersection of the sets A and B
$A \cup B$	$\{x \mid x \in A$ and $x \in B\}$, the union of the sets A and B
$\emptyset$	The **empty set** (the set with no elements), also known as the **null set**

Remark A.1.1. Note that Munkres uses the notation $A - B$ for set complement, but we will not be following that convention.

The set of real numbers is denoted by $\mathbb{R}$. The set of **integers** is a subset of real numbers, given by

$$\mathbb{Z} := \{\dots, -2, -1, 0, 1, 2, \dots\}.$$

We may use subscripts to indicate certain subsets of $\mathbb{Z}$, such as

$$\mathbb{Z}_{>0} := \{a \in \mathbb{Z} \mid a > 0\}.$$

The set of **natural numbers** is the subset of the integers given by

$$\mathbb{N} := \{0, 1, 2, \dots\}.$$

Be warned that some mathematicians actually choose to not include 0 in the natural numbers. Finally, the **rational numbers** is the set $\mathbb{Q}$ of all real numbers that can be written in the form $\frac{a}{b}$ with a and b integers and $b \neq 0$.

Here are some properties of sets:

Theorem A.1.2 (Algebra of set operations)**.** *Let A, B, and C be subsets of some universe U. Then the following hold:*

Properties of $\emptyset$ and the universe	$A \cap \emptyset = \emptyset$ $A \cup \emptyset = A$	$A \cap U = A$ $A \cup U = U$
Idempotent Laws	$A \cap A = A$	$A \cup A = A$
Commutative Laws	$A \cap B = B \cap A$	$A \cup B = B \cup A$
Associative Laws	$(A \cap B) \cap C = A \cap (B \cap C)$ $(A \cup B) \cup C = A \cup (B \cup C)$	

Theorem A.1.3 (Properties of the complement)**.** *Let A and B be subsets of some universe U. With U fixed, we write $A^c := U \setminus A$. Then the following hold:*

Basic properties	$(A^c)^c = A$ $A \setminus B = A \cap B^c$	
Empty set and universe	$A \setminus \emptyset = A$ $A \setminus U = \emptyset$	$\emptyset^c = U$ $U^C = \emptyset$
Subsets and complements	$A \subseteq B \iff B^c \subseteq A^c$	

Theorem A.1.4 (Distributive Laws)**.** *Given a set A and sets B_α index by a set Λ,*

$$A \cap \left(\bigcup_{\alpha \in \Lambda} B_\alpha \right) \quad \textit{and} \quad A \cup \left(\bigcap_{\alpha} B_\alpha \right) = \bigcap_{\alpha \in \Lambda} A \cup B_\alpha.$$

Theorem A.1.5 (De Morgan's Laws)**.** *Given a set A and sets B_α index by a set Λ,*

$$A \setminus \left(\bigcup_{\alpha \in \Lambda} B_\alpha \right) = \bigcap_{\alpha \in \Lambda} (A \setminus B_\alpha) \quad \textit{and} \quad A \setminus \left(\bigcap_{\alpha} B_\alpha \right) = \bigcup_{\alpha \in \Lambda} (A \setminus B_\alpha).$$

Definition A.1.6. Given a set A, the **power set** of A is the set $\mathcal{P}(A)$ of all subsets of A.

Exercise A.1.7. Show that if A is a finite set with n elements, then $\mathcal{P}(A)$ is a finite set with 2^n elements.

Note that if A is infinite then $\mathcal{P}(A)$ is infinite.

Definition A.1.8. Let A be a set.

- We say that A is **finite** if A is empty or there exists an integer $n \geqslant 1$ and a bijection $f \colon A \longrightarrow \{1, \dots, n\}$.
- We say that A is **countably infinite** if there is a bijection $f \colon A \longrightarrow \mathbb{Z}_{>0}$.
- We say that A **countable** if it is finite or countably infinite.
- We say that A is **uncountable** if it is not countable.

A.2 Functions

Definition A.2.1. A **function** from a set A to a set B is a rule that associates to each element of $x \in A$ exactly one element of the set B. We usually give functions a name, and denote the function f from A to B by $f \colon A \to B$. The function $f \colon A \to B$ has three components:

- The set A is called the **domain** of f.
- The set B is called the **codomain** of f.
- The rule that associates elements from the domain with elements of the codomain.

In particular, note that while we may refer to the function as simply "the function f", indicating a rule without indicating the domain and codomain does not define a function.

Definition A.2.2. Let $f \colon A \to B$ be a function with domain A and codomain B. Given $a \in A$, we write $f(a)$ for the image of a under f, meaning the element of B that f associates to a. If $f(a) = b$, then a is called a **preimage** of b. Given a subset $C \subseteq A$, the **image** of C under f is

$$f(C) \coloneqq \{f(c) \mid c \in C\}.$$

The **image** of f is the set

$$f(A) \coloneqq \{f(a) \mid a \in A\}.$$

The image of f is also sometimes denoted by $\mathrm{im}(f)$ or $\mathrm{image}(f)$, and it is sometimes instead called the **range** of f.

Given a subset $C \subseteq B$, the **preimage** of C under f is

$$f^{-1}(C) = \{a \in A \mid f(a) \in C\}.$$

Remark A.2.3. Given a function $f\colon A \longrightarrow B$, note that the image of f is a subset of the codomain B of f. Moreover, the preimage of any subset C of B is a subset of the domain A.

Definition A.2.4. Given a set A, the **identity function** on A is the function $\mathrm{id}_A\colon A \longrightarrow A$ given by $\mathrm{id}_A(a) = a$ for all $a \in A$. Given sets $A \subseteq B$, the **inclusion** of A into B is the function $\iota\colon A \longrightarrow B$ given by $\iota(a) = a$ for all $a \in A$.

Definition A.2.5. Let $f\colon A \longrightarrow B$ be a function.

- We say f is **injective** if
$$f(x) = f(y) \implies x = y.$$
Equivalently, $|f^{-1}(\{b\})| \leqslant 1$ for every $b \in B$.
- We say f is **surjective** if $f(A) = B$, or equivalently, if for all $b \in B$ there exists $a \in A$ such that $f(a) = b$. Equivalently, $f^{-1}(\{b\})$ is nonempty for every $b \in B$.
- We say f is **bijective** if f is both injective and surjective.

Exercise A.2.6. Let $f\colon A \longrightarrow B$ be a function.

a) Show that f is injective if and only if there exists a function $g\colon B \longrightarrow A$ such that $gf = \mathrm{id}_A$.

b) Show that f is surjective if and only if there exists a function $g\colon B \longrightarrow A$ such that $fg = \mathrm{id}_B$.

c) Show that f is bijective if and only if f has an inverse, meaning that there exists a function $g\colon B \longrightarrow A$ such that $fg = \mathrm{id}_B$ and $gf = \mathrm{id}_A$.

Notation A.2.7. If $f\colon A \longrightarrow B$ is a bijective function, we write f^{-1} to indicate the unique function $f^{-1}\colon B \longrightarrow A$ satisfying $ff^{-1} = \mathrm{id}_B$ and $f^{-1}f = \mathrm{id}_A$.

Definition A.2.8. Given a collection of sets $\{A_\alpha\}_{\alpha\in\Lambda}$ indexed by a set Λ and a fixed $\beta \in \Lambda$, the **projection map** $\pi_\beta\colon \prod_{\alpha\in\Lambda} A_\alpha \longrightarrow A_\beta$ is the function given by
$$\pi_\beta((a_\alpha)_{\alpha\in\Lambda}) = a_\beta.$$

Definition A.2.9. The **product function** of $f_\alpha\colon A \longrightarrow B_\alpha$ with $\alpha \in I$ is the function
$$(f_\alpha)_{\alpha\in I}\colon A \longrightarrow \prod_{\alpha\in I} B_\alpha$$
given by
$$(f_\alpha)_{\alpha\in I}(a) = (f_\alpha(a))_{\alpha\in I}.$$

Lemma A.2.10. *Consider sets A and B_α for each $\alpha \in I$.*

a) For every function $f\colon A \longrightarrow B_\alpha$ there exist functions $f_\alpha\colon A \longrightarrow B_\alpha$ such that
$$f = (f_\alpha)_{\alpha\in I} \quad \text{and} \quad \pi_\beta \circ f = f_\beta \text{ for all } \beta \in I.$$

b) Given functions $f_\alpha\colon A \longrightarrow B_\alpha$, the product function $f = (f_\alpha)_{\alpha\in I}$ satisfies
$$\pi_\beta \circ f = f_\beta \text{ for all } \beta \in I.$$

A.3 Cartesian products

Given a collection of sets, there are multiple ways we can use them to create a new set.

Definition A.3.1. Given two sets A and B, the **cartesian product**, or more simply the **product** of A and B is the set of **ordered pairs**

$$A \times B := \{(a, b) \mid a \in A, b \in B\}.$$

More generally, given a collection of sets A_α indexed by a set I, their cartesian product is the set

$$\prod_{\alpha \in I} A_\alpha = \{(a_\alpha)_{\alpha \in I} \mid a_\alpha \in A_\alpha \text{ for all } \alpha \in I\}$$

of tuples (a_α), where a tuple (a_α) corresponds to an element $a_\alpha \in A_\alpha$ for each α.

It is a bit obnoxious to define this formally, but here is the proper definition: the Cartesian product of $\{A_\alpha\}_{\alpha \in I}$ is the set of all functions

$$f \colon I \longrightarrow \bigcup_{\alpha \in I} A_\alpha$$

such that $f(\alpha) \in A_\alpha$ for all $\alpha \in I$. Each such function is an element in the Cartesian product; this corresponds precisely to picking out an element $a_\alpha \in A_\alpha$ for each α.

Remark A.3.2. Let A and J be any sets. The cartesian product of A with itself indexed by J, which we write as

$$A^J := \prod_{\alpha \in J} A$$

as the set of all functions from J to A:

$$A^J = \{f \colon J \longrightarrow A \text{ any function}\}.$$

Indeed, note that such a function f corresponds to a choice of $f(\alpha) \in A$ for each $\alpha \in J$.

Remark A.3.3. One can then write the **Axiom of Choice** as a statement about Cartesian products:

$$\text{If } A_\alpha \neq \emptyset \text{ for every } \alpha \in I, \text{ then}$$

$$\prod_{\alpha \in I} A_\alpha \neq \emptyset.$$

We now record some useful properties of cartesian products.

Theorem A.3.4. *The following hold for all sets A, B, and C:*

a) $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

b) $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

c) $(A \cap B) \times C = A \times C \cap B \times C$.

d) $(A \cup B) \times C = A \times C \cup B \times C$.

e) $A \times (B \setminus C) = (A \times B) \setminus (A \times C)$.

f) $(A \setminus B) \times C = (A \times C) \setminus (B \times C)$.

g) If $C \subseteq A$, *then* $C \times B \subseteq A \times B$.

h) If $C \subseteq B$, *then* $A \times C \subseteq A \times B$.

Exercise A.3.5. Show that

$$A \times B \cap U \times V = (A \cap U) \times (B \cap V).$$

A.4 Equivalence relations and order relations

Definition A.4.1. A **relation** on a set A is a subset of the cartesian product $A \times A$. We sometimes denote the relation by a symbol, such as $\sim$, and write $a \sim b$ to indicate that the pair (a, b) is an element of the relation. We may also write $a \nsim b$ to indicate that (a, b) is not in the relation.

Definition A.4.2. An **equivalence relation** on a set X is a relation $\sim$ satisfying the following properties:

- Reflexivity: $x \sim x$ for all $x \in X$.
- Symmetry: if $x \sim y$ then $y \sim x$.
- Transitivity: if $x \sim y$ and $y \sim z$, then $x \sim z$.

Given an equivalence relation $\sim$ on X, the **equivalence class** of $x \in X$ is the set

$$[x] \coloneqq \{y \mid y \sim x\}.$$

An equivalence class is any subset of X of the form $[x]$ for some $x \in X$.

Remark A.4.3. Any equivalence relation on a set X partitions X by equivalence classes: every element of X is in some equivalence class, and any two distinct equivalence classes are disjoint. Thus we can write X as a disjoint union of equivalence classes:

$$X = \bigcup_{x \in X} [x].$$

We will often think of the resulting set of equivalence classes. Informally, one may think about starting from a set X and *glueing* elements together to make each equivalence class, forming a new (potentially smaller) set.

Conversely, given any partition of X into disjoint nonempty subsets

$$X = \bigcup_{\alpha \in I} Y_\alpha,$$

we can define an equivalence relation by setting $x \sim y$ if x and y are in the same subset Y_α. The subsets Y_α are thus the equivalence classes for $\sim$. In summary: to give an equivalence relation on X is equivalent to giving a partition of X.

Notation A.4.4. Given an equivalence relation $\sim$ on a set X, we denote the set of all equivalence classes by $X/\sim$. The **canonical surjection** is the map $p\colon X \longrightarrow X/\sim$ given by $x \mapsto [x]$.

Example A.4.5. Let $f\colon X \longrightarrow Y$ be any surjective function. Consider the relation $x \sim y$ given by $f(x) = f(y)$. Then $\sim$ is an equivalence relation (exercise!). The equivalence classes are the **fibers**

$$f^{-1}(\{y\}) = \{x \in X \mid f(x) = y\}$$

for $y \in Y$.

Note that the relation $\sim$ is an equivalence relation even if f is not onto, but we need f to be onto for all the fibers to be nonempty.

Abstractly, this is the *only* example: all equivalence relations arise in this way.

Lemma A.4.6. *Let $\sim$ be an equivalence relation on a set X, and consider the canonical projection map $p\colon X \longrightarrow X/\sim$. The equivalence relation induced by the surjective map p is $\sim$.*

Proof. Let $\sim_p$ denote the equivalence relation induced by p. Given $x, y \in X$,

$$x \sim_p y \iff p(x) = p(y) \iff [x] = [y] \iff x \sim y. \qquad \square$$

Definition A.4.7. Let $f\colon X \longrightarrow Y$ be a function, and $y \in Y$. The **fiber** of y under f is the preimage of y under f, that is, $f^{-1}(\{y\})$. The **fibers** of f are all the sets of the form $f^{-1}(\{y\})$.

Remark A.4.8. Let $f\colon X \longrightarrow Y$ be a function. Under the equivalence relation induced by f, that is, such that $x \sim y$ when $f(x) = f(y)$, the fibers of f are precisely the equivalence classes.

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Nomenclature

$(a, b]$	$\{x \in X \mid a < x \leqslant b\}$
(a, b)	$\{x \in X \mid a < x < b\}$
$[a, b)$	$\{x \in X \mid a \leqslant x < b\}$
$[a, b)$	$\{x \in X \mid m \leqslant x \leqslant b\}$
$\mathscr{B}_<$	basis for the order topology
$\prod_{\alpha \in I} A_\alpha$	cartesian product of the sets A_α, where $\alpha \in I$
$\overline{S}$	closure of S
$\overline{S}^X$	closure of S in X
$\mathbb{R}^\omega$	$\prod_{n \in \mathbb{Z}} \mathbb{R}$
$\mathbb{R}_\ell$	$\mathbb{R}$ with the lower limit topology
$\mathbb{R}_K$	$\mathbb{R}$ with K-topology
$]a, b[$	$\{x \in X \mid a < x < b\}$
$B_\varepsilon^d(x)$	$\{y \in X \mid d(x, y) < \varepsilon\}$
S'	accumulation points of S
S^{int}	interior of S
$X/\sim$	set of equivalence classes for the equivalence relation $\sim$ on X

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